Tata Memorial Centre

Center For Cancer Epidemiology

AFFORDABLE CANCER CARE for One and All



TRAINING MANUAL FOR DOCTORS

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Basics of Cancer

Cancer

You must have often wondered, what is cancer? How is it caused? What can we do to prevent it? How is it diagnosed and treated? Well, this chapter is there to address these issues only. Having awareness about the cancer is the first step one can take in preventing it.

What is cancer?

Cancers are a group of diseases associated with abnormal growth of cells. Without any check, the disease may keep on progressing ultimately leading to pre-mature death. They can arise anywhere in the body and can affect people from all age groups, socio-economic strata and race. Cancer is the leading cause of morbidity and mortality in the world. According to data by International Agency for Research on Cancer, there were 141 lakh new cancer cases, 82 lakh cancer deaths and 326 lakh people living with cancer in 2012 worldwide. In our own country, about 4.7 lakh new cases of cancer are detected annually. Cancer is responsible for death of about 3.5 lakh people annually in India itself.

Lung cancer is the most common cause of death from cancer worldwide, estimated to be responsible for nearly one in five deaths (15.9 lakhs deaths, 19.4% of the total). Amongst women, breast cancer is the commonest cause of death and is responsible for 5.2 lakh deaths annually. These figures are just a numerical representation of the vast damage caused by cancer worldwide. One might want to assume that with recent rapid developments in medical sciences, the incidence and prevalence of cancer may be coming down. Sadly, that isn't so. According to WHO, within next two decades annual incidence of cancer may increase up to 220 lakhs. With increase in adoption of modern lifestyle with unhealthy habits like lack of physical activity, decreased intake of fruit and vegetables, increasing use of tobacco, fast food, alcohol, etc. the incidence of cancer is expected to further increase.

Cancer and its molecular basis

Cancer refers to unregulated and unrestricted proliferation* of cells. Clinically it is seen in the form of a growth. Neoplasm is an abnormal mass of tissue growing in an uncoordinated way and persisting even when the stimuli responsible for growth has been removed. A tumor is said to be benign when its characteristics are considered relatively harmless suggesting that it wouldn't spread to near-by or distant areas, can be operated easily and doesn't risk the patient's life significantly. Malignant tumors are collectively referred to as cancers, derived from the Latin word for crab, because they involve the tissues in a manner similar to a crab. Malignant tumor can invade and destroy adjacent structures and spread to distant sites (metastasize) to cause death. Not all cancers cause death, if detected early and treated properly, some cancers can be cured. Cancers result from certain changes at the molecular level in response to various external or internal stimuli.

- Physical carcinogens*- ultraviolet and ionizing radiation
- Chemical carcinogens- components of tobacco smoke, aflatoxin, asbestos, arsenic etc
- Biological carcinogens - viruses, bacteria or parasites

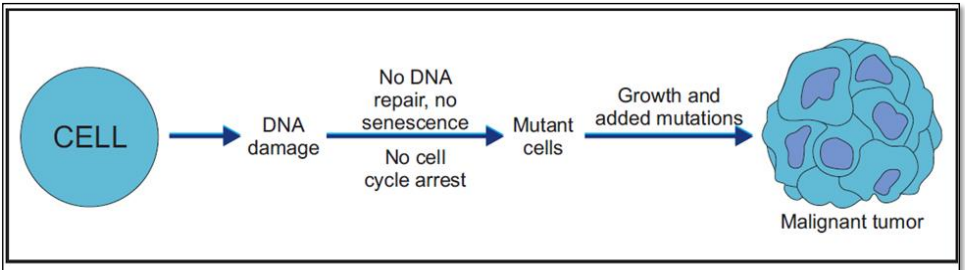
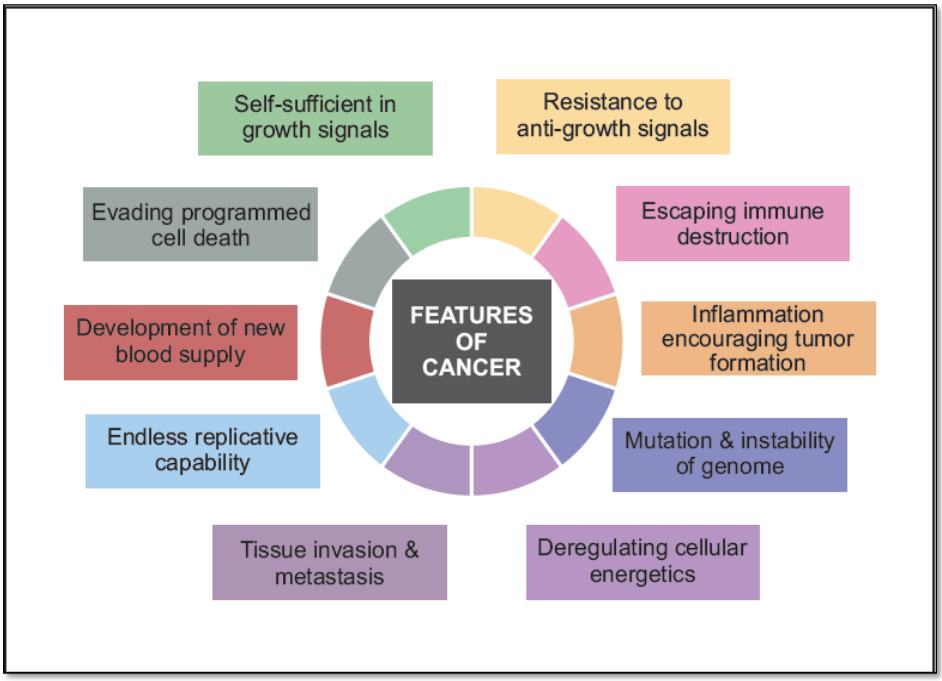
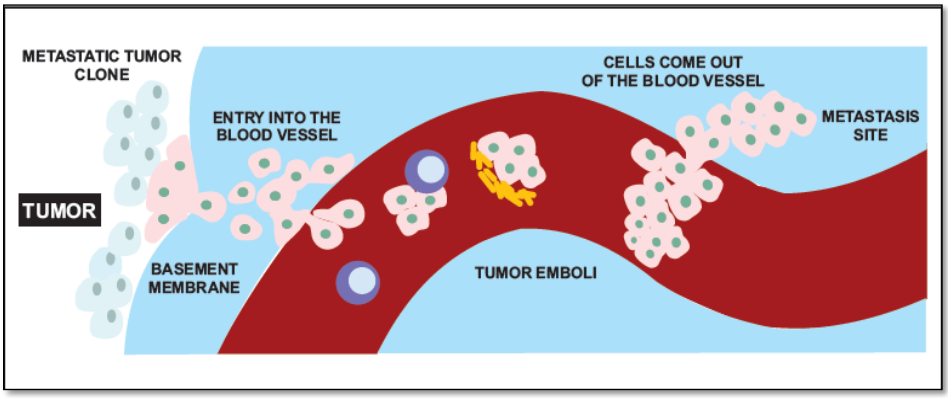


Figure 1: Simplified depiction of how a cell undergoes malignant change – with DNA damage, no repair and further mutations

Table 1: Differences between benign and malignant growth

	Malignant Growth	**Benign Growth**
Growth	Unregulated, fast growing	Slow growing
Surrounding Structures	Invaded or infiltrated	Compressed
Metastasis	Yes	No
Periphery	Ill-defined borders, capsule may be absent	Well-defined, Encapsulated





What are the common cancers of human body?

Worldwide, the commonest cancers amongst men are lung, prostate, colo-rectum, stomach, and liver cancer. Among women the most common sites diagnosed are breast, colo-rectum, lung, cervix, and stomach cancer. In India, amongst males common cancers in decreasing frequency are oral, lung, stomach and colo-rectal cancers. In females, breast, cervix, colo-rectal, ovary and oral cancers are the commonest cancers in decreasing frequency.

The incidence of these cancers varies according to the geographical area, prevalent social customs and the socio-economic strata. For example, oral cancers are common in Indian sub-continent and not that common in western countries. This is because of higher consumption of chewable forms of tobacco in the form of gutkha, paan, paan masala, khaini, supari etc. Cervical cancers are commoner in women from lower socio-economic strata because of poor genital hygiene. Colo-rectal cancers occur with higher frequency in those consuming more of fatty food and less of fibers in their diet.

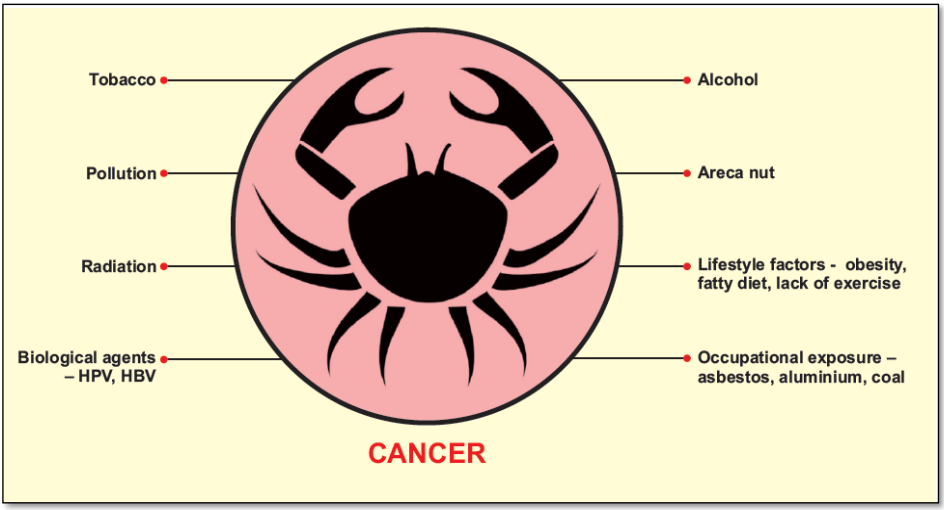
What are early signs of Cancers?

Early identification of cancer is essential for proper treatment. Cancer may present in a variety of ways. Ten common signs of cancers are:

- New lumps or growths or swelling;
- A sore or bruise that does not heal;
- A mole that changes in shape, size or colour or bleeds;
- Persistent cough or hoarseness that last;
- Indigestion or difficulty in swallowing;
- A change in bowel or bladder habits;
- Shortness of breath;
- Loss of appetite;
- Unexplained weight loss or tiredness;
- Blood in urine, bowel motions or sputum.
- Post coital bleeding or bleeding in post-menopausal women.

In case of presence of any of these symptoms, the person should approach the nearest doctor and get himself evaluated to rule out cancer.

Figure 4: Various causative factors associated with cancer



What are the common causes of cancers?

Common causes of cancer are as shown in Figure 4 and are as follows:

Tobacco - Tobacco consumption is the single most important avoidable risk factor for cancer mortality worldwide. According to WHO, it causes an estimated 22% of cancer deaths per year. Majority of the lung cancers are associated with smoking. Passive smoking has also been associated with cancers in non-smoking adults. Risk increases with increasing quantity of smoking. In Indian sub- continent use of smokeless tobacco is more prevalent. These include use of gutkha, pan masala, masheri, raw tobacco, betel quid etc. Tobacco is associated with cancers of lung, oral cavity, throat, oesophagus, urinary bladder, pancreas, kidney, liver, stomach, bowel, cervix, ovary, nose and sinus as well as some types of leukaemia (Figure 5). Tobacco contains over 4000 types of chemicals. Out of these, around 200 are harmful for human body and about70 different chemicals have been found to be carcinogenic (Figure 6). Various studies have shown that those who quit tobacco have a better survival than those who don't. About 50% of tobacco users die because of some form of tobacco related disease.

Over the counter preparations are available for nicotine replacement and these have been marketed as solution for tobacco addiction. The safety of such preparations is dubious. They have not been found to be that efficacious in curbing tobacco addiction. They themselves contain chemicals like nicotine which may cause cancer itself. A new trend, is that of the use of e-cigarettes. They are battery operated LED light containing devices which light up when used and release nicotine. They have been falsely advertised as safer alternative to cigarettes. Many studies have been conducted upon the safety of these e-cigarettes. They have been found to be associated with persistent throat irritation, seizures, pneumonia and heart failure. They contain nicotine which is highly addictive and is itself carcinogenic. Their fumes also have been found to have tobacco specific nitrosamines and

other harmful chemicals. Various countries have banned or restricted the sale of such devices.

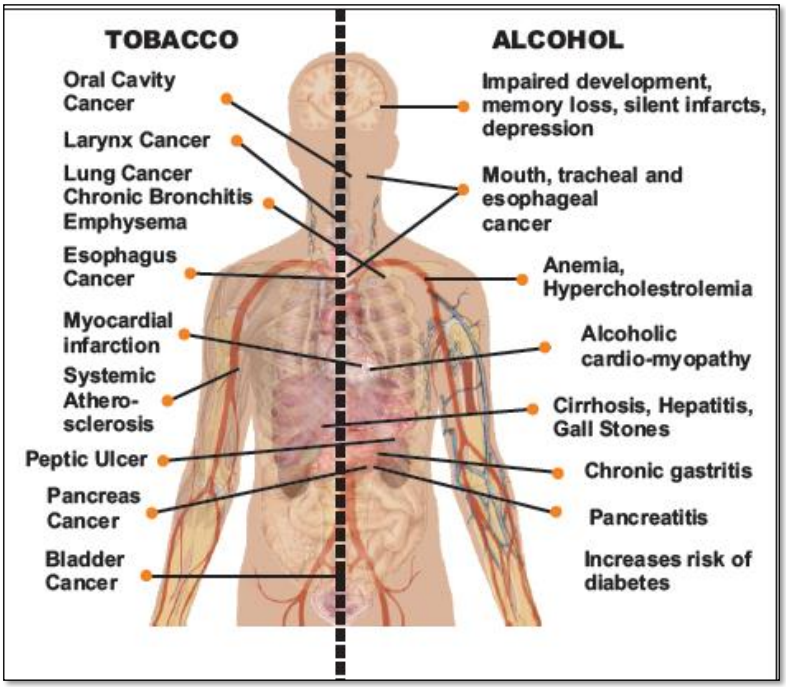


Figure 5: Common adverse effects of tobacco and alcohol

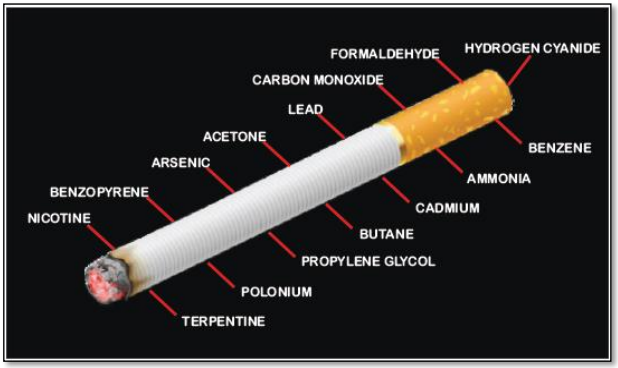


Figure 6: Harmful chemicals present in cigarette smoke

- Tobacco kills around 1 million people in
- India each year.
- Smoking causes 30% of all cancer deaths,
- 17% of all heart disease deaths and atleast
- 80% of deaths due from bronchitis and emphysema.
- 52.3% of adults were exposed to second- hand smoke at home.
- 29.0% of adults were exposed to second- hand smoke in public places

Alcohol: Alcohol itself is associated with several cancers, the risk of which increases with increasing quantity of alcohol consumed. It has synergistic action along with smoking and if a person consumes both alcohol and tobacco then the risk of developing cancer is much higher than the risk associated with consuming them independently. Alcohol use is a risk factor for many cancer types including cancer of the oral cavity, pharynx, larynx, oesophagus, liver, colo-rectum and breast (Figure 5).

Areca Nut: It is also called as supari in India. It may be chewed alone or in combination with betel leaf, catechu and slaked lime – called as Pan or betel quid. Powdered areca nut in ready to eat mixtures with other ingredients is called Pan Masala. If tobacco is added to these then it is called Gutkha. It is associated with Sub Mucous Fibrosis which is a pre-cancerous condition where the mouth opening of the chewer decreases gradually. It has been included as a group I carcinogen by The International Agency for Research into Cancer (IARC). It uses has been found to lead to oral cancer. It may also result in cancers of liver and pancreas.

Pollution: This includes the environmental pollution of air, water and soil with carcinogenic chemicals. Exposure to these chemicals can occur through air and drinking water. Chemicals like arsenic cause contamination of drinking water and can result in lung cancers. Household air is polluted by coal fires and cause lung cancer. Besides, contamination of food with aflatoxins can also cause cancer.

Lifestyle factors: These include unhealthy eating habits and lack of physical activity. High body mass index is associated with cancers of esophagus, colo- rectum, breast, endometrium etc. Excessive consumption of red meat is associated with colo- rectal cancers. Healthy diet, including vegetables, salads and fruits have been found to decrease risk of developing certain cancers.

Obesity: It refers to excessive fat accumulation. If body mass index (BMI) is above 25 the person is called overweight. Those having BMI over 30 are called obese. It is associated with increased risk of heart diseases, diabetes and cancer like those of endometrium, colon, breast, esophagus, pancreas etc. Besides increasing the risk of developing such cancers, obesity is also associated with increased chances of deaths due to such cancers. Proper healthy diet and regular exercise are required to curb the menace of obesity.

Occupational exposure: Several cancers are seen to have higher association with specific occupations. Substances like asbestos, cadmium, ethylene oxide, benzopyrene, silica, ionizing radiation including radon, tanning devices, aluminum and coal production, iron and steel founding, have been found to be associated with various cancers. These are important to know because with adequate knowledge and precautions most of these cancers can be prevented. Common cancers associated with occupational exposure include lung, bladder, leukemia, skin etc.

Radiation including UV rays: Ultraviolet radiation is associated with skin cancers like basal cell carcinoma, squamous cell carcinoma and melanoma*. Exposure to ionizing radiation has been associated with increased risk of several cancers. This has been extensively studied in survivors of atomic bomb in Japan.

Biological agents: These include various viruses, parasites and bacterial infections which predispose the patient to develop certain cancers. Human papilloma virus (HPV) is associated with cervical cancers and oropharyngeal cancers. Hepatitis B and C are associated with liver cancers. Parasitic

(Schistosomiasis) infection is associated with increased risk of bladder cancer. *H. pylori* infection may results in stomach cancer.These infections can be prevented with precaution, vaccination and early diagnosis and treatment.

Screening for cancer

Screening refers to identification of a disease in an individual who has not yet developed its signs and symptoms. In this, simple tests are conducted on general population or a high-risk group to identify the individuals harboring the disease. In this way, the affected individual can be identified before he/she presents with the symptoms of the disease. Screening helps in early identification and diagnosis of several cancers. Identifying these cancers early means a better possibility of control over the disease. Screening is a useful tool in early detection of cancers of breast, cervix, oral cavity and colo- rectum.

Breast cancer:

Screening for breast cancer is done with mammography. Mammography refers to taking an x-ray image of the breast. Its efficacy as a screening test has been proven in several studies and it helps in reducing the morbidity and mortality associated with breast cancer. It should be done regularly in individuals having family history of breast cancer. Self-examination of breast can also help the woman to become aware about swelling in breast.

Cervical cancer:

Pap smear is used to screen women for cervical cancer (Figure 7). In this a smear taken from the cervix is evaluated under microscope to look for pre-cancerous or cancer cells. It is a cost-effective method of screening. It is advisable for all women above the age of 35-40 years to undergo Pap smear examination. Besides this, visual inspection of the cervix with acetic acid application is also done at several places.

Oral cancers:

Screening for oral cavity cancers is relevant because of easy accessibility of oral mucosa for visual inspection without requiring any special equipment or expertise and the fact that most of the oral carcinomas are preceded by visible pre-cancerous lesions. Commonly seen pre-cancerous lesions include leukoplakia and erythroplakia (Figure 8). Leukoplakia refers to a whitish patch in oral cavity, for the presence of which no other cause can be found. A similar reddish patch is called erythroplakia. Biopsy from these suspicious areas may be taken to rule out malignancy



Figure 8: Clinical photograph of the oral cavity of a tobacco chewer, smoker showing pre-cancerous and cancerous lesion.

Colo-rectal Cancers:

Screening for these cancers is directed at people with family history of similar cancers or in those suffering from conditions like inflammatory bowel disease where there are higher chances of developing malignancy. Presence of occult blood (not visible grossly) is tested for in feces and in certain cases colonoscopy is also done.

How is cancer treated?

Clinical examination -A patient with cancer has to be evaluated with a thorough history of symptoms and clinical examination. Endeavour is made to assess the extent of the disease and its relationship with surrounding vital structures. If required, endoscopy may be required to assess the lesion. Presence of any other disease which may affect the treatment of cancer is also looked for. Presence of any distant metastases is also evaluated.

Radiologic assessment - Clinical examination is supplemented with radiologic imaging. This may include X-ray, Ultrasonography, CT scan, MRI scan or PET scan as clinically applicable. They help in better evaluation of the lesion, its extent and presence of loco-regional or distant metastasis.

Pathologic assessment – Along with clinical evaluation, a small part of the lesion is biopsied* and sent for histopathological*examination.

If required aspiration from the lesion is sent for cytological* examination. In both these tests, microscope and some special stains are used to assess the presence of and type of cancer cells.

Once proper clinico-radiological assessment has been done, the cancer is staged as per the existing staging guidelines. Further treatment will depend upon the type of cancer and its stage. Various modalities available are surgery, radiotherapy and chemotherapy. These may be used individually or in combination with each other as per disease status.

Surgery–Surgical excision* with reference to cancer refers to excision of the lesion with wide margins. Along with the primary lesion, regional lymph nodes are also addressed. The defect created may be

closed primarily or may have to be reconstructed with local, regional or free flaps.



Figure 9: Operation theatre – Surgery in progress

- **Biopsy: small part of disease is cut and sent for further examination**
- **Histopathological, cytological examination**
  - **methods of analyzing the cells under a microscope using different stains**
- **Excision – complete removal of the disease**

Radiotherapy – It involves the use of ionizing radiation for treatment of cancer. It may be used as a primary modality or as an additional therapy following or preceding surgery. Ionizing radiation acts by formation of free radicals which cause DNA damage and cell death. Radiation beam is shaped in such a way so as to cause minimum damage to surrounding normal tissues. When the source of radiation is present outside the body, it is called external beam radiotherapy (Figure 10). In some cases the source may be placed at the tumour site, in such cases it is called brachytherapy



Figure 10: A patient of cancer being treated by external beam radiotherapy using linear accelerator device



Figure 11: Chemotherapy involves administration of drugs to kill the cancer cells.

Chemotherapy – As mentioned in figure 11 a wide range of drugs are used to treat cancer. These drugs have a variety of action and act on different stages of cell cycle to cause cell death. The detail of their mechanism of action and activity is beyond the scope of this chapter. These may be used in different settings for cancer treatment. It may be the primary modality used along with radiation or may be used in adjuvant setting or also as a palliative therapy.

Palliative care – Many a time, cancer may progress beyond a stage where a curative treatment is possible. Patient may have a very advanced stage of or may have developed distant metastasis. In cases, where cure is not expected the patient is treated with a palliative intent. Palliative care involves relieving the symptoms of cancer rather than curing it. Chemotherapy or radiotherapy may be used in selective group of patients. If the general condition of the patient is very poor, then he is maintained on best supportive care. Medications are given to ease of the pain and the patient is helped to carry out activities of daily living. Hospice care (institutions providing end-of-life care) may also be required in some cases.

Prevention of cancer

First and foremost, comes – cessation of tobacco consumption. Tobacco is the single most important cause of deaths related to cancer in up to 20% of people. One should stay away from both chewable as well as non-chewable forms of tobacco.

Alcohol cessation – alcohol itself is carcinogenic and also has increased carcinogenic effect when consumed along with tobacco.

Healthy lifestyle – regular physical activity along with diet rich in fruits and vegetables help in reducing the chances of developing cancer.

Good genital hygiene – maintenance of good genital hygiene along with safe sexual practices reduce the risk of infection with Human Papilloma Virus (HPV) and decrease the risk of developing cervical and oropharyngeal cancer. Vaccination against HPV infection is also available for adolescent girls.

Curbing pollution - reduction of both household and outdoor pollution helps in reduction of risk of cancer. Household pollution with fossil fuels like coal can result in lung cancer. Outdoor pollution includes contamination of both air and water with various carcinogens.

Work place precautions – those involved in occupations associated with exposure to carcinogenic compounds or radiation should be provided adequate protection and should be examined at regular intervals.

Hepatitis B Virus infection can result in liver cancer. Proper precautions should be taken to prevent the spread of infection through infected blood products. Vaccination against Hepatitis B virus is also available and its full dose should be taken.

Family history – those with a family history of breast, thyroid, colo-rectal cancer should be aware of risk associated with their developing cancer. They should have themselves examined by a doctor and should be aware of the early signs of cancer.

Cancer Screening and Early Detection Guidelines

Overview

Malignancies (cancer) can be broadly classified into haematological and solid malignancies. Haematological malignancies (leukaemia's, lymphomas) apart from routine blood test and imaging, often needs advanced investigations like flow cytometry, FISH etc. Also, treatment options ranges from exhaustive lines of chemotherapies to bone marrow transplantation. Hence, they are best treated at tertiary centres by medical oncologist or haematologists, where dedicated ICU and other support systems like full-fledged blood bank are available.

The most common cancers in India are lung, oral cavity, breast, cervix, stomach, colorectal, oesophageal malignancies. More than 90% of lung cancers, present in very advanced stages and not amenable to cure. Oral cavity, breast and cervix form the major chunk of curable malignancies in India. These cancers can be screened effectively and if detected early cure rates are very high. These malignancies need simple chemotherapy regimens and select cases in them needs less radical surgeries which can be managed at some district hospitals or medical colleges. This will reduce the huge burden on existing cancer institutes and also will help the patients, to receive quality cancer care near their place of residence. This manual deals with an overview of screening of these three most common malignancies. Also, this manual covers topics on principles of cancer management which gives an overview of evaluation and treatment options for these most common cancers.







What is screening?

At the pre-cancer stage the patient will not have any complaints and only the screening tests can detect the disease. So, the screening test is done on apparently normal patient, irrespective of whether they have any symptoms or not. An ideal screening test would be one that is simple, painless, can be done rapidly on a large number of individuals, low cost and able to detect the disease accurately.

Protocol for Early Detection for Cervical Cancer

Cervical cancer is the most common cause of cancer deaths among women in India. Nearly 130,000 new cases of cervical cancer occur each year in our country and 80,000 women die annually from the disease. The advanced stage at first diagnosis (ranging from 70 – 95% in different centers) accounts for the poor survival of the cervical cancer patients in the country. Cancer of uterine cervix develops slowly over 10 to 15 years following oncogenic Human Papilloma Virus infection. Hence frequent screening in women can help in detecting cervical cancer early.

Risk factors for Cervical Cancer

Human papillomavirus (HPV) infection has been established as the necessary, but not solely sufficient, cause of cervical cancer. The vast majority of women infected with an oncogenic HPV type never develop cervical cancer, which suggests that additional factors acting in conjunction with HPV influence the risk of disease development. Cofactors such as parity, use of oral contraceptives, tobacco smoking, immunosuppression— particularly related to human immunodeficiency virus (HIV), infection with other sexually transmitted diseases, and poor nutrition all have been associated, to

various extents, with the development of invasive cervical cancer. Their specific role in the development of cervical cancer remains unclear, however. Age of sexual debut, lifetime number of sexual partners, history of sexually transmitted infections, and other characteristics of sexual activity are linked to the likelihood of becoming infected with HPV and are not considered to be cofactors for the progression from HPV infection to cervical cancer.

Gross Anatomy of the Reproductive Tract

The female reproductive system (or female genital system) contains two main parts: the uterus, which hosts the developing fetus, produces vaginal and uterine secretions, and can pass sperm through to the Fallopian tubes; and the ovaries, which produce the female's egg cells. These parts are internal; the vagina meets the external organs at the vulva, which includes the labia, clitoris and urinary meatus. The vagina is attached to the uterus through the cervix, while the uterus is attached to the ovaries via the Fallopian tubes. At certain intervals, the ovaries release an ovum, which passes through the Fallopian tube into the uterus. If, in this transit, it meets with sperm, a single sperm can enter and merge with the egg, fertilizing it. Corresponding equivalent among males is the male reproductive system.

During the reproductive process, the egg is not a passive recipient, but rather an active participant in the fertilization process. It releases certain molecules that are essential to guiding the sperm which allow the surface of the egg to attach to the sperm's surface. The egg can then absorb the sperm and fertilization begins. The fertilization usually occurs in the oviducts, but can happen in the uterus itself. A zygote will then divide over enough generations of cells to form a blastocyst, which implants itself in the wall of the uterus, where it begins the processes of embryogenesis and morphogenesis. When developed enough to survive outside the womb, the cervix dilates and contractions of the uterus propel the fetus through the birth canal, which is the vagina.

The ova are larger than sperm and have formed by the time a female is born. Approximately every month, a process of oogenesis matures one ovum to be sent down the Fallopian tube attached to its ovary in anticipation of fertilization. If not fertilized, this egg is flushed out of the system through menstruation.

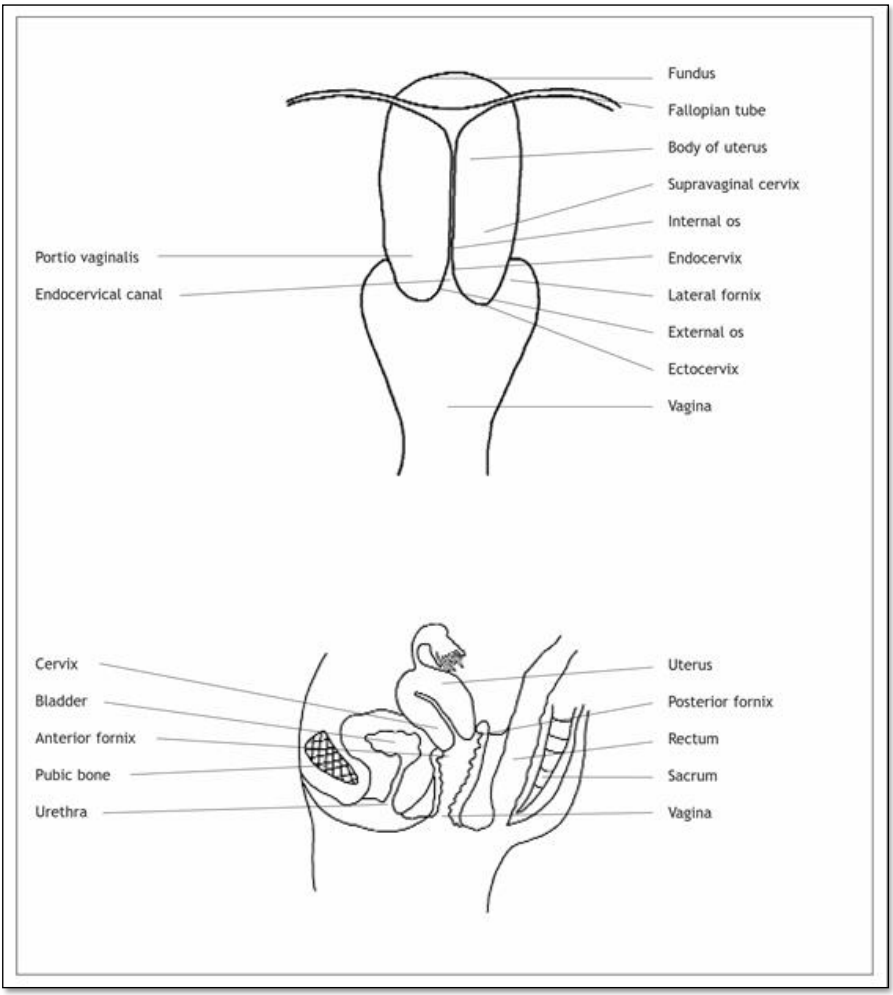


Figure 11: Gross Anatomy of the Reproductive Tract

Uterine Cervix: Brief Description

The cervix or cervix uteri (Latin: neck of the uterus) is the lower part of the uterus in the human female reproductive system. In a non-pregnant woman, the cervix is usually 2 to 3 cm long (~1 inch) and roughly cylindrical in shape. The narrow, central cervical canal runs along its entire length, connecting the uterine cavity and the lumen of the vagina. The opening into the uterus is called the internal os, and the opening into the vagina is called the external os. The lower part of the cervix, known as the vaginal portion of the cervix (or ectocervix), bulges into the top of the vagina.

The cervical canal is a passage through which sperm must travel to fertilize an egg cell after sexual intercourse. Several methods of contraception, including cervical caps and cervical diaphragms aim to block or prevent the passage of sperm through the cervical canal. Cervical mucus is used in several methods of fertility awareness, such as the Creighton model and Billings method, due to its changes in consistency throughout the menstrual period. During vaginal childbirth, the cervix must flatten and dilate to allow the fetus to progress along the birth canal. Midwives and doctors use the extent of the dilation of the cervix to assist decision-making during childbirth.

The endocervical canal is lined with a single layer of column-shaped cells, while the ectocervix is covered with multiple layers of cells topped with flat cells. The two types of epithelia meet the squamocolumnar junction. Infection with the human papillomavirus (HPV) can cause changes in the epithelium, which can lead to cancer of the cervix. Cervical cytology tests can often detect cervical cancer and its precursors, and enable early successful treatment. Ways to avoid HPV include avoiding sex, using condoms, and HPV vaccination. HPV vaccines, developed in the early 21st century, reduce the risk of cervical cancer by preventing infections from the main cancer-causing strains of HPV.

Symptoms of Cervical Cancer:

- Vaginal bleeding: This includes bleeding between periods, after sexual intercourse (post coital) or post-menopausal bleeding.
- Unusual vaginal discharge: A watery, pink or foul-smelling discharge is common.
- Pelvic pain: Pain during intercourse or at other times may be a sign of abnormal changes to the cervix.

All of these cervical cancer symptoms should be discussed with your doctor.

Signs of advanced stages of cervical cancer

Cervical cancer may spread (metastasize) within the pelvis, to the lymph nodes or elsewhere in the body. Signs of advanced cervical cancer include:

1. Back pain
2. Leg pain or swelling
3. Leakage of urine or faeces from the vagina
4. Weight loss
5. Fatigue

Protocol for Early Detection for Cervical cancer

Screening for cervical cancer can be considered in women aged 30 to 50 years, as the chances of detecting pre-cancerous lesions are maximum in this age group. Screening women at frequent intervals, as in developed countries, puts a heavy burden on the limited manpower and financial resources, in our country. Hence, at least, if we are able to achieve a good coverage (more than 70%) of the target women, we could reduce the number cervical cancer deaths. There are number of screening methods like Pap test, visual inspection with acetic acid (VIA), visual inspection with lugols iodine (VILI). Though pap`s test is standard screening test it requires more laboratory facilities and a trained cytopathologist, which is available only at some district hospitals or medical colleges.

Introduction to Cervical Intraepithelial Neoplasia CIN

Cervical intraepithelial neoplasia (CIN) is a premalignant lesion that is diagnosed by histology as CIN1, CIN2, or CIN3. If left untreated, CIN2 or CIN3 (collectively referred to as CIN2+) can progress to cervical cancer.

It is estimated that approximately 1–2% of women have CIN2+ each year, with higher rates reported for women of HIV-positive status, at 10%. A diagnosis of CIN2+ is a histological diagnosis obtained from biopsies of the suspect lesions, either with or without colposcopy, for which treatment is recommended. Adenocarcinoma in situ (AIS) is a precursor lesion for cervical cancer that is diagnosed by cytology and can be treated. The majority of AIS are found in the transformation zone. AIS may be associated with CIN. There are three principal treatments available in low- and middle-income countries to treat CIN: cryotherapy, large loop excision of the transformation zone (LLETZ, or LEEP), and cold knife conisation (CKC).

Colposcopy

A Colposcope is a low-power, stereoscopic, binocular field microscope with a powerful light source used for magnified visual examination of the uterine cervix to help in the diagnosis of cervical neoplasia.

Indications for colposcopy

- Suspicious-looking cervix
- Invasive carcinoma on cytology
- CIN 2 or CIN 3 on cytology
- Persisting (for more than 12-18 months) low-grade (CIN 1) abnormalities on cytology
- CIN 1 on cytology
- Persistently unsatisfactory quality on cytology
- Infection with oncogenic human papillomaviruses (HPV)
- Acetopositivity on visual inspection with acetic acid (VIA)

- Acetopositivity on visual inspection with acetic acid using magnification (VIAM)
- Positive on visual inspection with Lugol's iodine (VILI)

Table 2.1: Correlation between dysplasia/carcinoma in situ, cervical intraepithelial neoplasia (CIN) and the Bethesda terminology

Dysplasia terminology	Original CIN terminology	Modified CIN terminology	The Bethesda system (SIL) terminology (1991)
Normal	Normal	Normal	Within normal limits Benign cellular changes (infection or repair)
Atypia			ASCUS/AGUS
	Koilocytic atypia, flat condyloma, without epithelial changes	Low-grade CIN	LSIL
Mild dysplasia or mild dyskaryosis	CIN 1	Low-grade CIN	LSIL
Moderate dysplasia or moderate dyskaryosis	CIN 2	High-grade CIN	HSIL
Severe dysplasia or severe dyskaryosis	CIN 3	High-grade CIN	HSIL
Carcinoma *in-situ*	CIN 3	High-grade CIN	HSIL
Invasive carcinoma	Invasive carcinoma	Invasive carcinoma	Invasive carcinoma

CIN: cervical intraepithelial neoplasia; LSIL: Low-grade squamous intraepithelial lesion; HSIL: High-grade squamous intraepithelial lesion; ASCUS: Atypical squamous cells of undetermined significance; AGUS: Atypical glandular cells of undetermined significance

Colposcopy Procedure

Colposcopy is usually done by a gynaecologist, a family medicine physician, or a nurse practitioner who has been trained to do the test. If a biopsy is done, the sample will be looked at by a pathologist. Patient is put in lithotomy position. Speculum is inserted to examine and the cervix. Vinegar (acetic acid) or iodine (Lugol's solution) may be used on cervix to make abnormal areas more visible and are examined using colposcope. Photographs or videos of the vagina and cervix may be taken.

If areas of abnormal tissue are found on the cervix, cervical biopsy is done. Usually several samples are taken. The samples are looked at under a microscope for changes in the cells that may mean cancer may be present or is likely to develop. If bleeding occurs, Monsel's or silver nitrate swab may be used on the biopsy area to stop the bleeding.

If a sample of tissue is needed from the endocervical canal, endocervical curettage (ECC) will be done. Since the endocervical canal cannot be seen by the colposcope, a curette is gently put into the endocervical canal to take a sample. ECC takes less than a minute to do and may cause mild cramping. An ECC is not done during pregnancy. Colposcopy and a cervical biopsy usually take about 15 minutes.

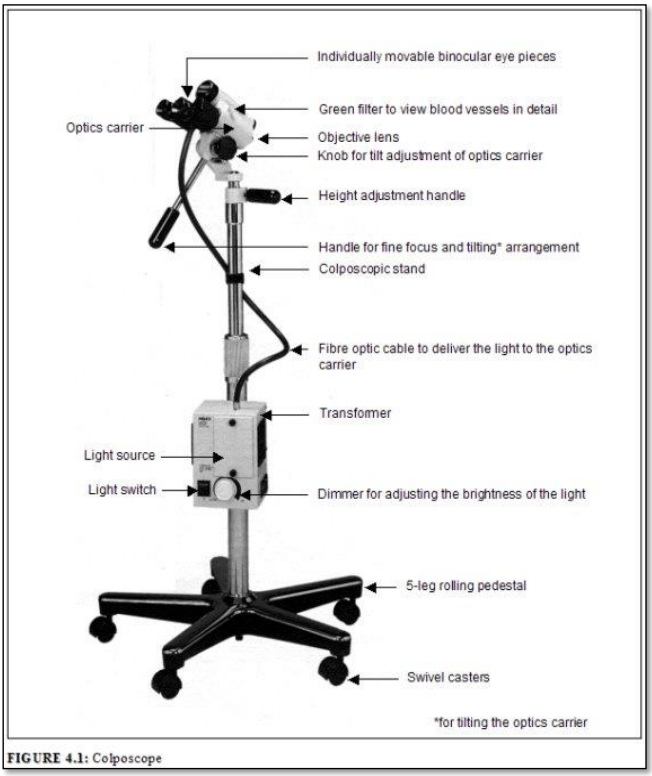


FIGURE 4.1: Colposcope

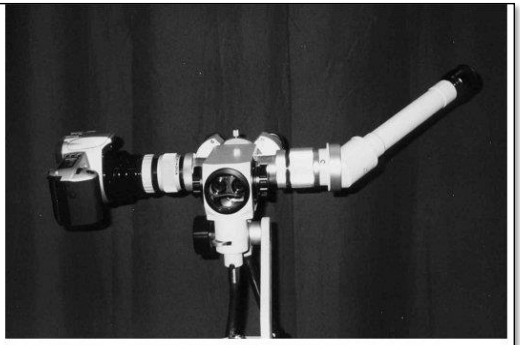


FIGURE 4.2: Colposcope with a photographic camera and a teaching side tube

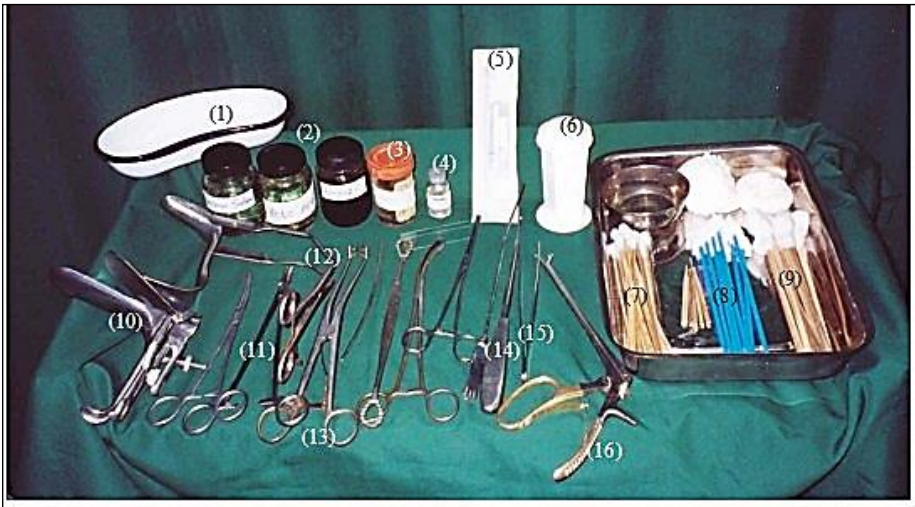


FIGURE 4.3: Colposcopy instrument tray

1: Kidney tray
2: Bottles with normal saline, 5% acetic acid and Lugol's iodine
3: Monsel's solution
4: Bottle containing formaline
5: Local anaesthetic syringe
6: Jar containing alcohol for cervical smear fixation
7: Cotton-tipped fine swab sticks
8: Cervical cytology brushes
9: Larger cotton-tipped swab sticks
10: Vaginal speculum
11: Sponge-holding forceps
12: Vaginal side-wall retractor
13: Endocervical speculum
14: Endocervical curette
15: Dissecting forceps
16: Punch biopsy forceps

Visual Inspection with Acetic acid (VIA) Test

VIA can be done at primary health centres and district hospitals. VIA can also be done at temporary screening camps like schools, community centres etc. It can be done by the nurses, female health workers or the GP`s at the primary health care level.

Instruments and other materials required to perform VIA Test

1. Sterile vaginal speculum
2. New examination gloves, or HLD surgical gloves
3. Large cotton swabs
4. Dilute (3-5%) acetic acid (vinegar) and a small bowl
5. Containers with 0.5% chlorine solution
6. A plastic bucket with a plastic bag
7. Quality assurance system to maximize accuracy

Preparation of 5% Acetic Acid

Ingredients

1. Glacial acetic acid: 5 ml
2. Distilled water: 95 ml

Preparation: Carefully add 5 ml of glacial acetic acid into 95 ml of distilled water and mix thoroughly.

Storage

Unused acetic acid should be discarded at the end of the day.

Label

5% dilute acetic acid

Note: It is important to remember to dilute the glacial acetic acid, since the undiluted strength causes a severe chemical burn if applied to the epithelium.

Steps for performing the VIA test

VIA involves naked eye examination (without magnification) of the uterine cervix after application of freshly prepared 3-5% acetic acid for a minute, under a good light source. The test results are scored as positive when a well-defined, dense acetowhite area with regular margins appears attached to the squamo columnar junction. The test is reported as positive if an acetowhite area is seen in or near the transformation zone or negative if no change is observed or suspicious for invasive cancer. (Squamo-columnar junction is identified as the pink smooth squamous epithelium joining the red velvet like columnar epithelium.)

Table 7.1: Surface extent of acetowhite areas associated with cervical neoplasia

Cervical neoplasia	Cases	One lip of cervix (%)	Both lips (%)
CIN 1	27	21 (78)	6 (22)
CIN 2	30	17 (57)	13 (43)
CIN 3	87	36 (41)	51 (59)
Early invasive cancer	66	10 (15)	56 (85)

Adapted from Burghart *et al.*, 1998

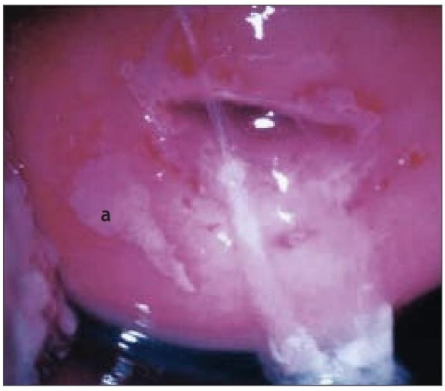


FIGURE 7.9: Geographic satellite lesion after application of 5% acetic acid (a) far away from the squamocolumnar junction, suggestive of low-grade lesion

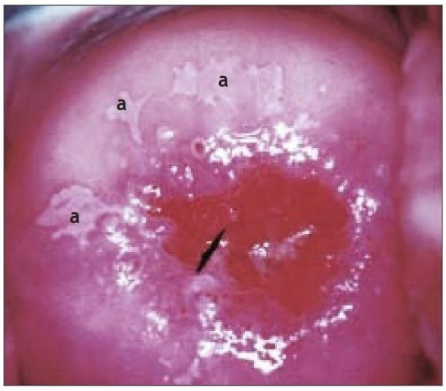


FIGURE 7.10: Geographic satellite lesions after application of 5% acetic acid (a) far away from the squamocolumnar junction, suggestive of low-grade lesions



FIGURE 7.11: Thin acetowhite lesion with geographic margins in the upper lip. Histology indicated CIN 1



FIGURE 7.13: Mildly dense acetowhite lesions arising from the squamocolumnar junction in 12 and 6 o'clock position with irregular geographical margins, which on histology proved to be CIN 1 lesion

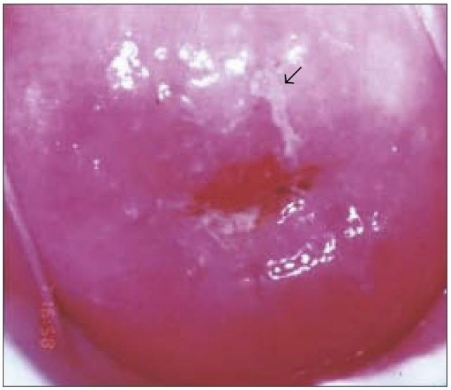


FIGURE 7.12: Mildly dense, thin, elongated acetowhite lesion with regular margins abutting the squamocolumnar junction. Note the fine mosaic at the distal end of the lesion. Histology indicated CIN 1

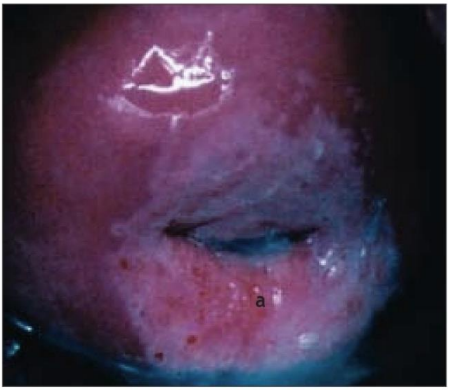


FIGURE 7.14: Note the circumorificial acetowhite CIN 1 lesion with irregular margin and fine mosaics (a)

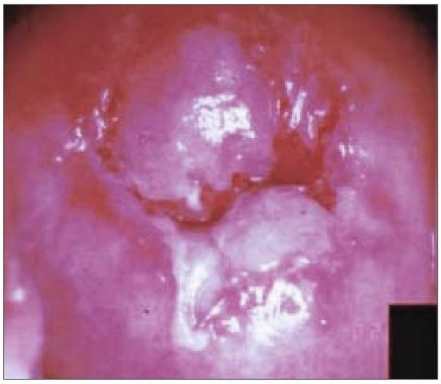


FIGURE 7.15: Moderately dense acetowhite lesions with irregular margins in the anterior and posterior lips (CIN 1)



FIGURE 7.17: Moderately dense acetowhite lesions with well defined margins and coarse punctations in the anterior lip and in 3 o'clock position (CIN 2 lesion)



FIGURE 7.16: Circumorificial, mild to dense acetowhite lesion with fine mosaic (arrow). Histology indicated CIN 1. Note the internal borders within the lesion (a)

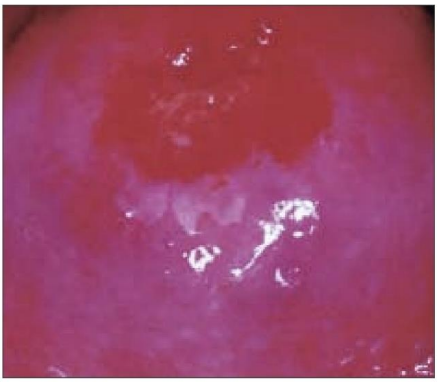


FIGURE 7.18: Dense, well defined acetowhite area with regular margins and coarse mosaic (CIN 2 lesion)

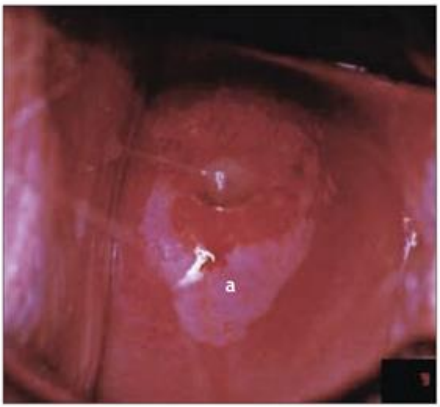


FIGURE 7.19: A dense acetowhite lesion with varying colour intensity and coarse mosaics (a) in a CIN 2 lesion



FIGURE 7.21: An acetowhite lesion arising at 12 o'clock position, abutting the squamocolumnar junction. Note the two colour intensities in the same lesion (a and b) with an internal border within the same lesion (c). This is an example of a lesion within a lesion

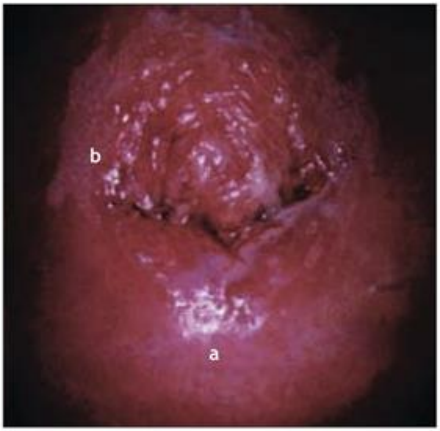


FIGURE 7.20: Acetowhite lesions with coarse punctation (a) and mosaics (b) in a CIN 2 lesion

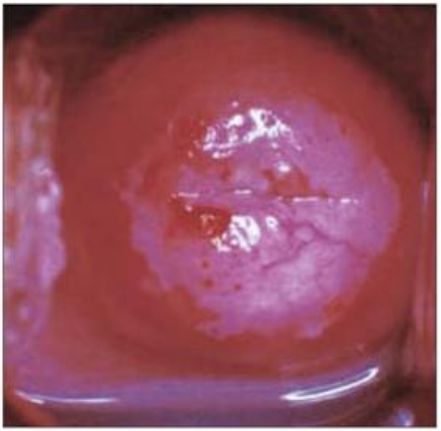


FIGURE 7.22: A circumorificial dense opaque acetowhite area with coarse mosaics (CIN 3 lesion)



FIGURE 7.23: A dense acetowhite lesion with regular margin and coarse, irregular punctation in a CIN 3 lesion

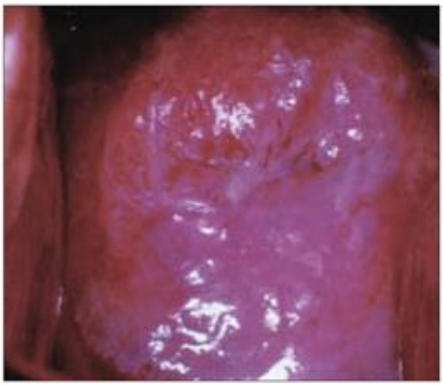


FIGURE 7.25: Note the intensely dense, complex, acetowhite lesion (CIN 3 lesion) with raised and rolled out margins, obliterating the external os



FIGURE 7.24: Coarse mosaics (a) in a CIN 3 lesion



FIGURE 7.26: A dense acetowhite lesion with raised and rolled out margins with a cuffed crypt opening (dense arrow) and coarse mosaics with umblication, suggestive of a CIN 3 lesion

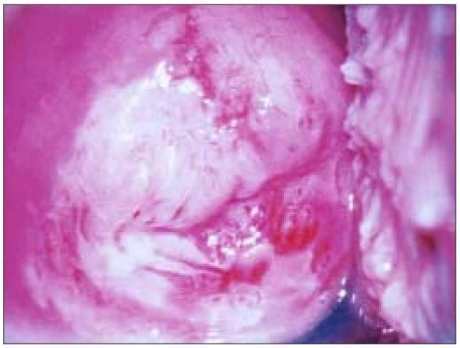


FIGURE 7.27: A dense acetowhite, opaque, complex, circumorificial CIN 3 lesion

Criteria for categorizing VIA test results

VIA Category	Description
Negative	• No aceto-white lesions • Transparent lesions or faint patchy lesions without definite margins • Nabothian cysts becoming aceto-white • Faint line like aceto-whitening at the junction of columnar and squamous epithelium • Aceto-white lesions far away from the transformation zone
Positive	• Distinct, opaque aceto-white area • Margin should be well defined, may or may not be raised • Abnormality close to the squamocolumnar junction in the transformation zone and not far away from the os.
Invasive cancer	• Obvious growth or ulcer in the cervix. Acetowhite area may not be visible because of bleeding.

Suspected cases would be referred to Medical Colleges & RCCs for further tests e.g. Pap Smear test, Colposcopy, biopsy etc. for confirmatory diagnosis of Cervical Cancer.

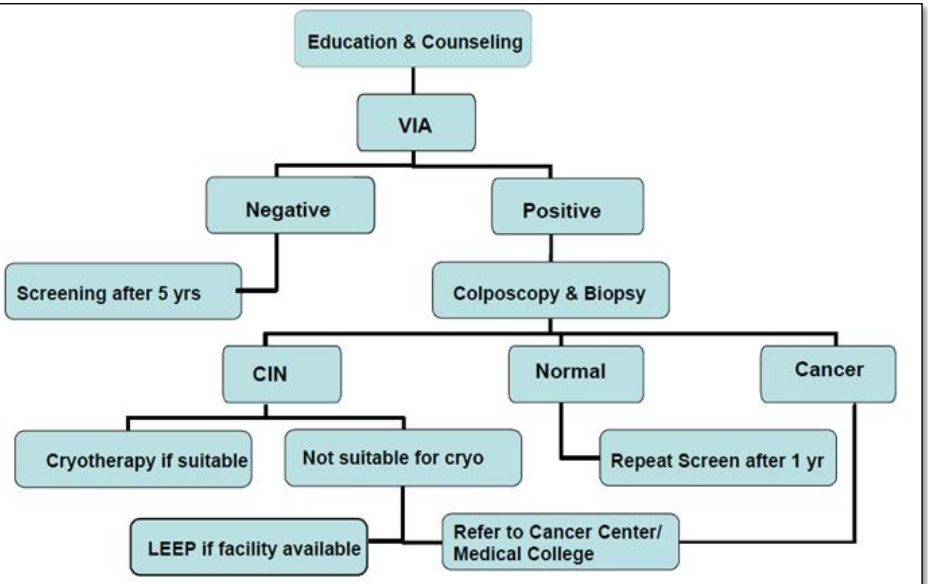


Figure 2: VIA based screening and management protocol flowchart

Pap Smear

Pap Smear is the standard screening test for cervical cancer. It is an easy and effective method revealing the presence of pre-cancerous lesions as well as in situ or very early invasive cancer. Though collection can be done by a trained health care worker, it involves a trained cytopathologist reporting. Hence this test is possible only at some district hospitals, medical colleges or regional cancer centers.

How is it done?

Patient lies on an examination table, a health care professional inserts a speculum into vagina. The health care provider then collects a sample of cells from the cervix by scraping it with a wooden /plastic spatula. An endocervical brush is rotated in the central opening of the cervix. There are two types of pap`s test. In a conventional Pap test, the specimen (or smear) is placed on a glass microscope slide and a fixative is added. In an automated liquid-based Pap cytology test, cervical cells collected with a brush or other instrument are placed in a vial of liquid preservative. The advantage of this method is it helps in HPV testing also. The samples are sent to pathologist for reporting.

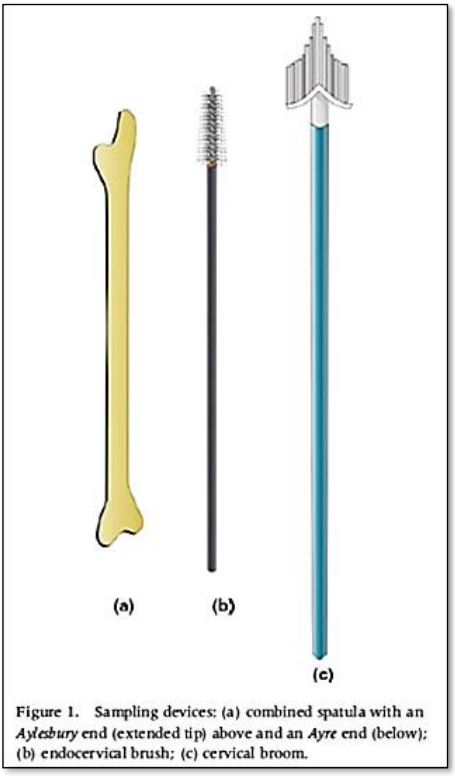
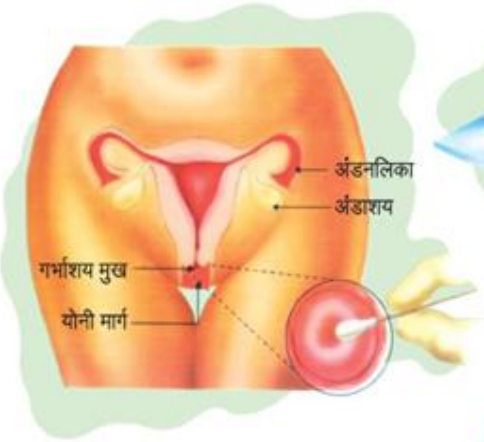
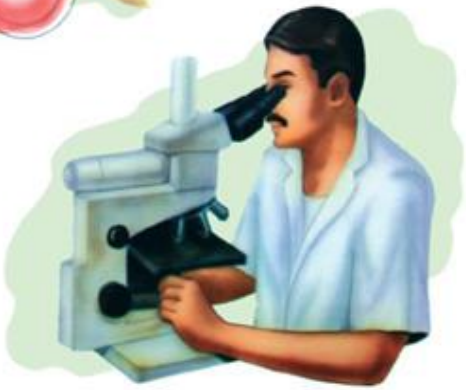


Figure 1. Sampling devices: (a) combined spatula with an *Ayledsbury* end (extended tip) above and an *Ayre* end (below); (b) endocervical brush; (c) cervical broom.




अंडनलिका
अंडाशय
गर्भाशय मुख
योनी मार्ग

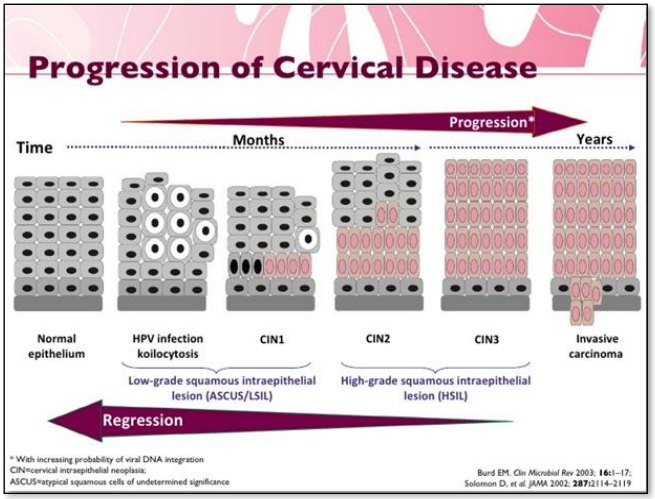




Abnormal results are reported according to the Bethesda system. They include:

- Squamous cell abnormalities (SIL)
  - Atypical squamous cells of undetermined significance (ASC-US)
  - Atypical squamous cells – cannot exclude HSIL (ASC-H)
  - Low-grade squamous intraepithelial lesion (LGSIL or LSIL)
  - High-grade squamous intraepithelial lesion (HGSIL or HSIL)
  - Squamous cell carcinoma
- Glandular epithelial cell abnormalities
- Atypical Glandular Cells not otherwise specified (AGC or AGC-NOS)

Endocervical and endometrial abnormalities can also be detected, as can a number of infectious processes, including yeast, herpes simplex virus and trichomoniasis. Any abnormal results in pap `s test is referred to colposcopy and biopsies and further treatment accordingly at medical colleges or RCC`s.



Management of Cervical Pre-cancers

Surgical treatments for precancerous cervical lesions, also known as cervical intraepithelial neoplasia (CIN), include both ablation and excision (Table 1). Ablative modalities are solely for treatment, while excisional methods provide a tissue specimen for histological assessment to define the severity of disease and the margins of abnormal tissue, for ensuring completeness of treatment. With ablative methods, biopsy specimens must be taken before treatment, if they are desired.

Surgical treatments for precancerous cervical lesions

Excision	Ablation
LEEP/LLETZ	Laser vaporization (CO2)
Electrosurgical needle conization	Cryotherapy
Laser conization (CO2)	CryoPen
Cold-knife conization	Cold coagulation
Hysterectomy	Electrocoagulation
	Electrocautery

Cryotherapy

Cryotherapy, which uses a low temperature probe to freeze abnormal cells, often is considered the most practical ablative method for use in low-resource settings. It is simple, inexpensive, and does not require electricity. Twelve months after treatment, cryotherapy is approximately 90 percent effective in treating HSIL.

Cryotherapy generally produces a lower cure rate for larger lesions (bigger than the tip of the cryotherapy probe or occupying, on average, over 75 percent of the surface area of the cervix) and for lesions that extend into the cervical canal. Where available, an alternative treatment plan should be considered for women with these types of lesions.

Cryotherapy Equipment and Component

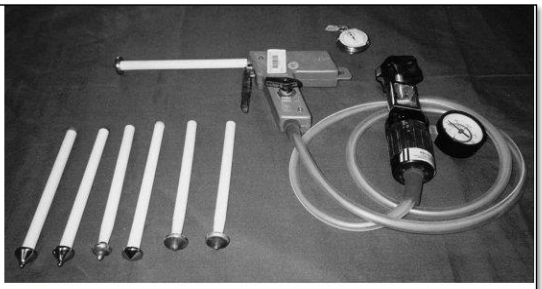


FIGURE 12.1: Cryoprobes, the cryogun, pressure gauge and stop watch.

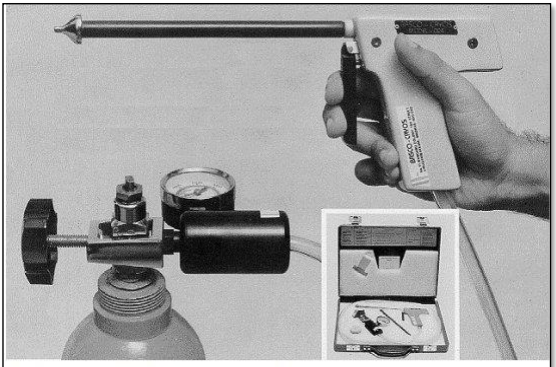
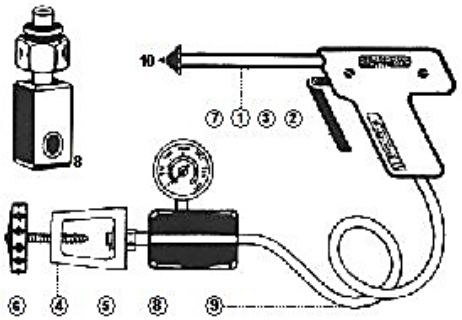


FIGURE 12.2: Cryotherapy equipment

The cryotherapy unit consists of a compressed gas cylinder (tank), a yoke with a tightening knob and an inlet of gas to connect the gas cylinder to the cryotherapy gun through a flexible gas-conveying tube, a pressure gauge showing the cylinder gas pressure, an outlet silencer, a cryotherapy gun with handle grip, a gas trigger to allow the gas to be released to the cryotherapy probe at high pressure and the cryotherapy probe. In most equipment, the pressure gauge shows three colour zones: yellow, green and red. When the gas cylinder is opened, if the pressure indicator in the gauge moves to the green zone, there is adequate gas pressure for treatment; if the needle remains in the yellow zone, the pressure is too low and the gas cylinder should be changed before commencing treatment; if the needle moves to the red zone, excess pressure is indicated and this excess pressure should be released. One should consult the manual provided by the manufacturer thoroughly for operational instructions.



1. Probe
2. Trigger
3. Handle grip (fibreglass)
4. Yoke
5. Instrument inlet of gas from cylinder
6. Tightening knob
7. Pressure gauge showing cylinder pressure
8. Silencer (outlet)
9. Gas-conveying tube
10. Probe tip

FIGURE 12.3: Components of cryotherapy equipment.

The step-by-step approach to cryotherapy

The physician or the nurse should explain the treatment procedure to the woman and reassure her. This is important to help the woman to relax during the procedure. After ensuring she has emptied her bladder, she should be placed in a modified lithotomy position and the cervix should be exposed with the largest speculum that can be introduced comfortably. The secretions are removed with a cotton swab soaked in saline. Then 5% acetic acid is applied and the cervix is examined with the colposcope. Following this, Lugol's iodine is applied to delineate the limits of the lesion. There is no need for local anaesthesia when performing cryotherapy.

The cryoprobe surface is wiped with saline to ensure adequate thermal contact with the cervix and optimal lowering of the tissue temperature. The cryotherapy probe tip is then firmly applied, with the centre of the tip on the os. It is obligatory to ensure that the vaginal walls are not in contact with the cryoprobe tip. The timer is then set and the gas trigger in the cryogun is released or squeezed to cool the cryoprobe in contact with the cervix. The gas escapes through the pressure gauge with a hissing noise. One should be able to observe ice being formed on the tip of the cryoprobe and on the cervix as freezing progresses. Make sure that the probe adequately covers the lesion and the tip does not

inadvertently contact and freeze any part of the vagina during the procedure.

Cryotherapy should consist of two sequential freeze-thaw cycles, each cycle consisting of 3 minutes of freezing followed by 5 minutes of thawing (3 minutes freeze-5 minutes thaw-3 minutes freeze-thaw). The treatment time should be monitored using a stop watch. Adequate freezing has been achieved when the margin of the ice ball extends 4-5 mm past the outer edge of the cryotip. This will ensure that cryonecrosis occurs down to at least 5 mm depth. To achieve this effect evenly throughout the treatment field, it is extremely important to establish and maintain excellent contact between the probe tip and the ectocervical surface. Once the second freeze for 3 minutes is completed, allow time for adequate thawing before removing the probe from the cervix. When thawing is completed, the ice formation on the cryoprobe tip is totally cleared and the probe is removed by gently rotating on the cervix. Do not attempt to remove the probe tip from the cervix until complete thawing has occurred. After removing the probe, examine the cervix for any bleeding.

Note the iceball formed in the cervix. The vagina should not be packed with gauze or cotton after cryotherapy to allow the secretions to escape. Women may be provided with a supply of sanitary pads to prevent the secretions staining their clothes.

After use, the probe tip should be wiped with 60-90% ethyl or isopropyl alcohol and then cleaned well with boiled water and disinfected with 2% glutaradehyde (see Chapter 14) and kept dry. After the procedure is completed, the cryogun, tubing, pressure gauge and gas tank should be de-contaminated by wiping with cotton soaked with 60-90% ethyl or isopropyl alcohol.

Follow-up after cryotherapy

Women should receive instructions on self-care and what symptoms to expect after treatment. They should be informed that they may experience some mild cramps and a clear or lightly blood-stained watery discharge for up to 4-6 weeks after treatment. Women should be advised not to use a vaginal douche or tampons or to have sexual intercourse for one month after treatment. They should be instructed to report if they have any one of the following symptoms in the six weeks after treatment: fever for more than two days, severe lower abdominal pain, foul-smelling- pus coloured discharge, bleeding with clots or bleeding for over two days. It is preferable to give written instructions on the above aspects and on follow-up.

Healing takes place during the first six weeks after cryotherapy. Granulation tissue is present in the wound during the first 2-3 weeks after cryotherapy which is followed by re-epithelialization of the surface. Normally, the wound is totally healed within 6-8 weeks of treatment. The appearance of the cervix 3 months and 12 months after cryotherapy.

The effect of cryotherapy on the potential transmissibility of human immunodeficiency virus (HIV) infection (to or from women) during the healing phase is not known. HIV-1 shedding in the vaginal secretions after treatment of CIN in HIV-positive women has been demonstrated. Therefore, the authors suggest advising all women that cryotherapy may increase the transmissibility of HIV and that using condoms is an effective means of prevention. Condoms should be used for a period of at least four but preferably six weeks. Ideally, a supply of condoms should be available free of charge at colposcopy clinics in settings where HIV infection is endemic.

Appointments should be made for a follow-up visit 9-12 months after treatment. During the follow-up, cytology and/or VIA should be performed, followed by colposcopy and directed biopsy depending upon the colposcopy findings, to assess the regression or persistence of lesions. Retreatment is carried out if lesions persist. Women who are

negative for neoplasia may be referred back to a screening programme (if one exists) or advised to undergo follow-up after three or five years.

Adverse effects, complications, and long-term sequelae

Cryotherapy is usually a painless procedure, if women have been properly reassured, their co-operation is obtained, and the procedure is carried out properly. Some women may experience some lower abdominal pain or cramps during and after cryotherapy. Once in a while, a woman may faint due to a vasovagal reaction. In such a situation, there is no need for panic and the women may be revived easily. Bleeding is extremely rare after cryotherapy.

Treated women experience a watery vaginal discharge for about 3-4 weeks after treatment. Vaginal bleeding is extremely unusual; it may be more likely to occur if freezing has been too aggressive and the ice ball has extended well past 5 mm in depth. The risk of post-operative infection is very slight and can probably be reduced further by delaying cryotherapy until any woman with a likely diagnosis of pelvic inflammatory disease (PID), sexually transmitted cervicitis (e.g., chlamydia or gonorrhea), vaginal trichomoniasis or bacterial vaginosis has been adequately treated and recovered. If a woman presents post-operatively with a malodorous discharge, pelvic pain and fever, the discharge may be cultured if possible, and empirical treatment should be prescribed with antibiotics that are effective for PID. Sexual partners should also be treated if the woman is diagnosed with PID, sexually transmitted cervicitis, or trichomoniasis. In developing countries, one may consider providing presumptive treatment with antibiotics routinely after cryotherapy (doxycycline 100 mg orally, two times a day, for seven days and metronidazole 400 mg orally, three times a day, for seven days).

Cervical stenosis occurs in less than 1% of women; reduced mucus production occurs in 5-10% of women. Cryotherapy has no known adverse effect on fertility and pregnancy. Invasive cancer has rarely been reported after cryotherapy, it is usually due to missed diagnosis as a result of poor diagnostic workup before cryotherapy.

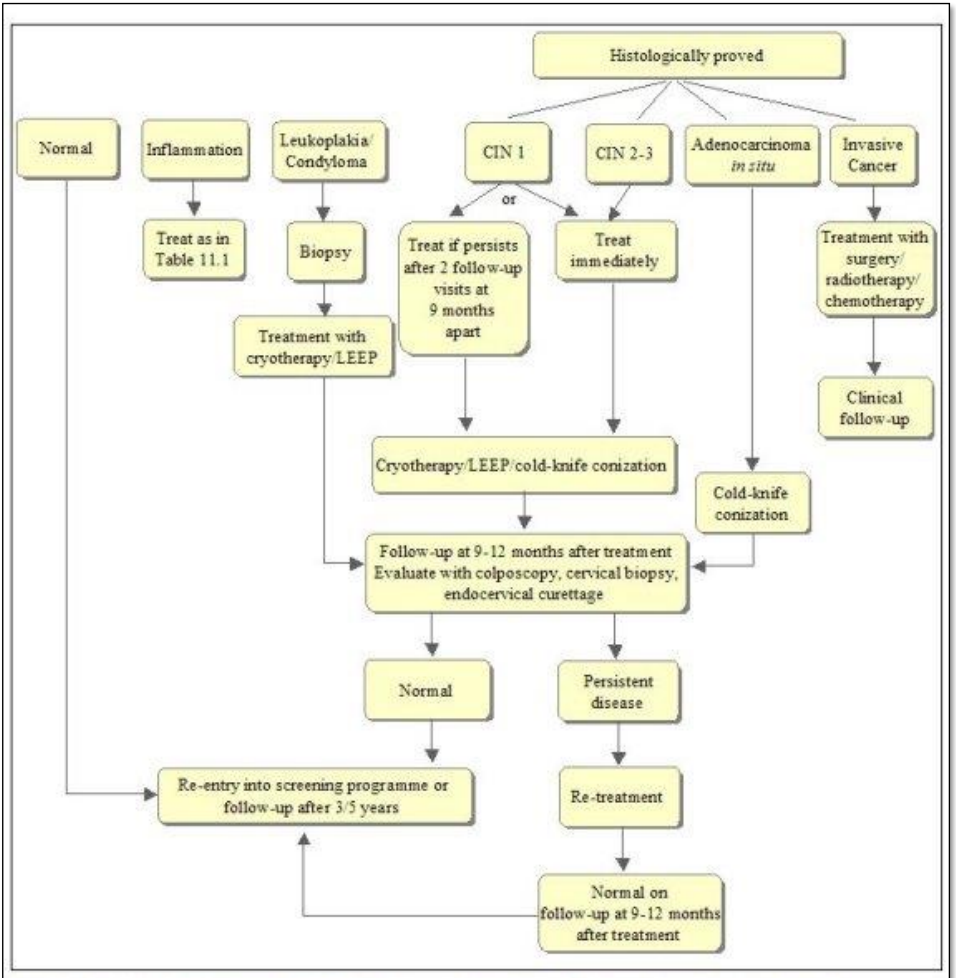


FIGURE 11.1: Flow chart of management decisions in cervical neoplasia and other conditions in low-resource settings. (CIN-cervical intraepithelial neoplasia; LEEP-Loop electrosurgical excision procedure)

Flow chart of management decisions in cervical neoplasia and other conditions in low-resource settings. (CIN-cervical intraepithelial neoplasia; LEEP-Loop electrosurgical excision procedure)

Treatment of Cervical Intraepithelial Neoplasia by Loop Electrosurgical Excision Procedure (LEEP)

Electrosurgery is the use of radiofrequency electric current to cut tissue or achieve haemostasis. A loop electrosurgical excision procedure (LEEP) operator needs to keep in mind that electricity flows to ground along the path of the least electrical resistance. The electrical energy used in electrosurgery is transformed into heat and light energy. The heat from a high-voltage electrical arc between the operating electrode and tissue allows the practitioner to cut by vaporizing tissue (at 100oC) or to coagulate by dehydrating tissue (above 100oC). The cutting electrodes are loops of very fine (0.2 mm) stainless steel or tungsten wire to achieve different widths, depths, and configurations of cut.

The higher temperatures involved in coagulation produce thermal effects greater than in electro-surgical cutting. This is important in electrosurgery, since an adequate pathological examination requires that the coagulation effect be minimal in the excised surgical specimen. On the other hand, some coagulation effect is desirable, even while cutting, in order to minimize bleeding in the surgical field. Manufacturers of modern electrosurgical generators are aware of the need to control bleeding. They offer electrosurgical cutting settings that lead to some coagulation by blending electrical currents, one with a cutting waveform and another with a coagulation waveform. This combination is called a blended cutting waveform, and is the type of waveform that will be referred to in this manual when electrosurgical cutting is discussed.

When the coagulation setting is selected on the electrosurgical generator, the coagulation waveform has a higher peak-to-peak voltage (producing higher temperatures) than that used for a pure cutting waveform, and is meant only to heat the tissue above 100°C to achieve dehydration. There are three types of coagulation: desiccation, in which the active electrode touches the tissue; fulguration, in which the active electrode does not touch the tissue but ‘sprays’ multiple sparks between itself and the tissue; and puncture coagulation, in which an electrode,

usually a needle, is inserted into the centre of a lesion. Coagulation using the fulguration setting and a 3- to 5-mm ball electrode is the type of coagulation that is normally referred to in this manual (one exception is be the use of a needle electrode to fulgurate a stubborn area of bleeding). The fulguration setting uses a higher peak-to-peak voltage waveform than the other coagulation settings, coagulating tissue with less current and, therefore, less potential harm to adjacent tissue.

To obtain a proper effect, an electrosurgical generator requires that a patient-return electrode or dispersive plate be used to allow the electrical circuit to be completed and the optimal current to flow. The dispersive plate should always be placed as close to the surgical site as possible. This is in contrast to the desired effect at the active electrode, where the current density is purposely high to concentrate the electrical energy as it is transformed into heat. Unless good electrical contact is maintained at the dispersive electrode over a large area, there is a danger that the patient will suffer from an electrical burn at this site. To guard against this possibility, modern electrosurgical units have circuitry (commonly referred to as a return electrode monitoring system) that continuously monitors the adequacy of the ground plate (dispersive pad) connection to the patient. This type of circuitry not only alerts the operator of a problem, but also prevents operation until the circuit fault is corrected. It is highly recommended that any electrosurgical generator meet the basic standards described above to ensure that safe and effective electrosurgery can be performed. It is assumed that only a system that meets or exceeds such requirements will be used in any of the electrosurgical procedures described in this manual.

Electrosurgery must not be performed in the presence of flammable gases, flammable anaesthetics, flammable liquids (e.g., alcohol-containing skin-preparation solutions or tinctures), flammable objects, oxidizing agents, or an oxygen-enriched atmosphere. The operator is, of course, at risk of receiving a burn from the active electrode if it is accidentally touched while activated.

The eligibility criteria that must be met before LEEP is performed

CIN is confirmed by cervical biopsy, when possible

If the lesion involves or extends into the endocervical canal, the distal or cranial limit of the lesion should be seen; the furthest (distal) extent is no more than 1 cm in depth

There is no evidence of invasive cancer or glandular dysplasia

There is no evidence of pelvic inflammatory disease (PID), cervicitis, vaginal trichomoniasis, bacterial vaginosis, anogenital ulcer or bleeding disorder

If the woman has recently delivered, she should be at least three months post-partum

Women with hypertension should have their blood pressure well controlled

The woman must give written consent to have the treatment after being thoroughly informed as to how it is performed and the probabilities of its effectiveness, adverse effects, complications, long-term sequelae, and alternative ways that are available to manage her problem

Eligibility criteria for cryotherapy

The entire lesion is located in the ectocervix without extension to the vagina and/or endocervix

The lesion is visible in its entire extent and does not extend more than 2 to 3 mm into the canal

The lesion can be adequately covered by the largest available cryotherapy probe (2.5cm) the lesion extends less than 2 mm beyond the cryotherapy probe

CIN is confirmed by cervical biopsy/colposcopy

There is no evidence of invasive cancer

The endocervical canal is normal and there is no suggestion of glandular dysplasia

The woman is not pregnant

If the woman has recently delivered, she is at least three months post-partum

There is no evidence of pelvic inflammatory disease

The woman has given informed written consent to have the treatment

Protocol for Early Detection for Breast Cancer

According to a study by the International Agency for Research on Cancer (IARC), there will be approximately 250,000 new cases annually of breast cancer in India by 2015. At present, India reports around 100,000 new cases annually according to the ICMR. About 30,000 women die from breast cancer in India annually. According to the ICMR, 1 out of 22 women in India is diagnosed with breast cancer and <3% of Indian women undergo screening for breast.

Regular breast screening is the best way to detect early breast cancer as they are most often successfully treated thus help saving lives, and it makes breast conservation surgery possible.

Anatomy of Breast

The breast is a mass of glandular, fatty and connective tissue. The breast is made up of:

- lobules – glands that produce milk
- ducts – tubes that carry milk from the lobules to the nipple
- fatty and connective tissue – surrounds and protects the ducts and lobules and gives shape to the breast
- areola – the pink or brown, circular area around the nipple that contains small sweat glands, which release (secrete) moisture as a lubricant during breast-feeding
- nipple – the area at the centre of the areola where the milk comes out

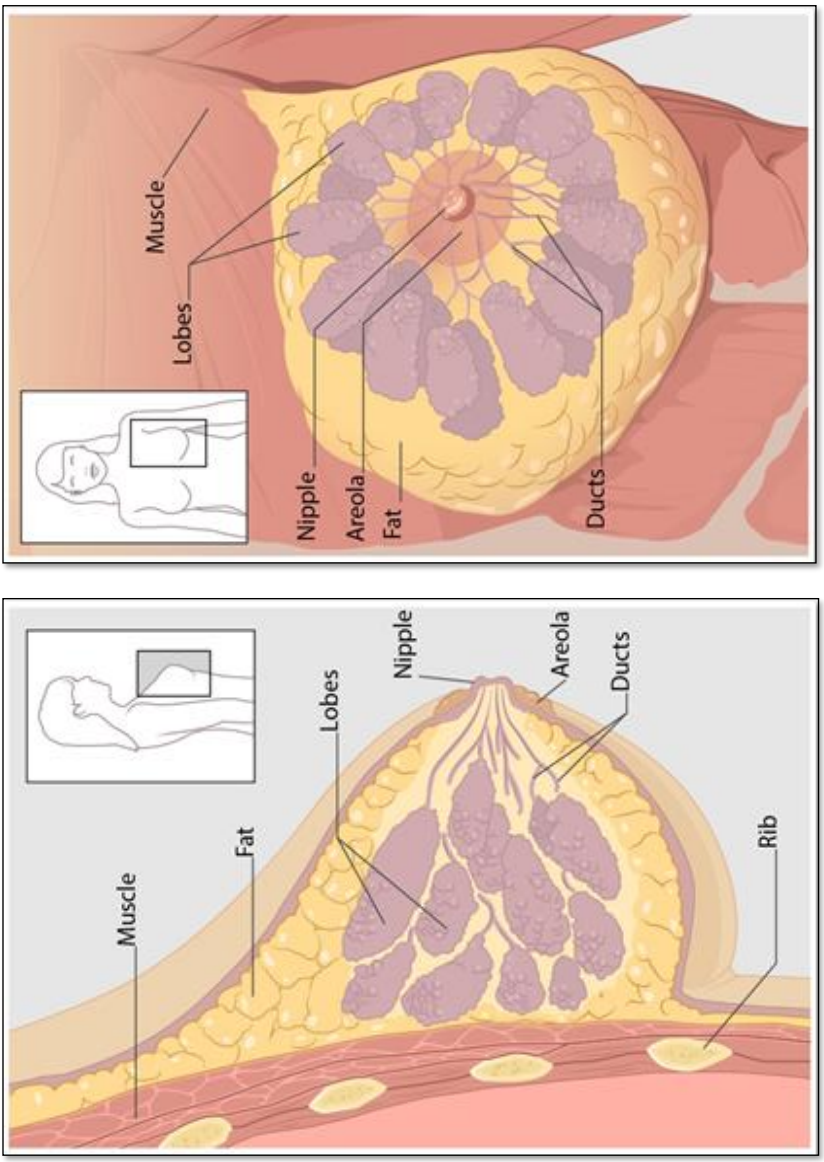


Fig: The anterior view and cross-section view of the breast.

Risk factors of Breast Cancer

- Getting older. The risk for breast cancer increases with age; most breast cancers are diagnosed after age 50.
- Genetic mutations. Inherited changes (mutations) to certain genes, such as BRCA1 and BRCA2. Women who have inherited these genetic changes are at higher risk of breast and ovarian cancer.
- Early menstrual period. Women who start their periods before age 12 are exposed to hormones longer, raising the risk for breast cancer by a small amount.
- Late or no pregnancy. Having the first pregnancy after age 30 and never having a full-term pregnancy can raise breast cancer risk.
- Starting menopause after age 55. Like starting one's period early, being exposed to oestrogen hormones for a longer time later in life also raises the risk of breast cancer.
- Not being physically active. Women who are not physically active have a higher risk of getting breast cancer.
- Being overweight or obese after menopause. Older women who are overweight or obese have a higher risk of getting breast cancer than those at a normal weight.
- Having dense breasts. Dense breasts have more connective tissue than fatty tissue, which can sometimes make it hard to see tumours on a mammogram. Women with dense breasts are more likely to get breast cancer.
- Using combination hormone therapy. Taking hormones to replace missing oestrogen and progesterone in menopause for more than five years raises the risk for breast cancer. The hormones that have been shown to increase risk are oestrogen and progestin when taken together.
- Taking oral contraceptives (birth control pills). Certain forms of oral contraceptive pills have been found to raise breast cancer risk.
- Personal history of breast cancer. Women who have had breast cancer are more likely to get breast cancer a second time.

- Personal history of certain non-cancerous breast diseases. Some non-cancerous breast diseases such as atypical hyperplasia or lobular carcinoma in situ are associated with a higher risk of getting breast cancer.
- Family history of breast cancer. A woman's risk for breast cancer is higher if she has a mother, sister, or daughter (first-degree relative) or multiple family members on either her mother's or father's side of the family who have had breast cancer. Having a first-degree male relative with breast cancer also raises a woman's risk.
- Previous treatment using radiation therapy. Women who had radiation therapy to the chest or breasts (like for treatment of Hodgkin's lymphoma) before age 30 have a higher risk of getting breast cancer later in life.
- Women who took the drug diethylstilbestrol (DES), which was given to some pregnant women in the United States between 1940 and 1971 to prevent miscarriage, have a higher risk. Women whose mothers took DES while pregnant with them are also at risk.
- Drinking alcohol. Studies show that a woman's risk for breast cancer increases with the more alcohol she drinks.

Methods of Breast Screening

1. Breast awareness

Breast awareness implies familiarity with one's own breast. Knowing what is normal for you may help you see or feel changes in your breasts. A breast self-examination (BSE) can be done monthly during the bath, best time being just at the end of menses. This helps to keep in notice any irregularity or any lumps in the breast.

A clinical breast exam (CBE) is a physical exam done by a trained medical staff. Steps of BSE and CBE explained later.

Warning signs of breast cancer



- Lump, hard knot or thickening inside the breast or underarm area
- Swelling, warmth, redness or darkening of the breast
- Change in the size or shape of the breast
- Dimpling or puckering of the skin
- Itchy, scaly sore or rash on the nipple
- Pulling in of your nipple or other parts of the breast
- Nipple discharge that starts suddenly
- New pain in one spot that does not go away.

Any woman with the above signs and symptoms should undergo CBE by health care professionals. The attending doctor should ensure prompt referral to higher center for imaging and FNAC of lump. However, patients should be reassured that all lumps are not cancer.

2. Mammography

A mammogram is an X-ray of the breast. Mammograms help to find breast cancer early, when it is easier to treat and before it is big enough to feel or cause symptoms. Regular mammograms once in 2 – 3yrs starting from the age of 50 yrs is advisable. But availability and cost are the limitations of this test.

Breast self-examination (BSE)

Best time to do Breast self-examination (BSE):

- Once a month
- Within a week after period ends.
- If not menstruating, person should pick a certain day-such as the first day of each month.
- If taking hormones, then do it 1-2 days after withdrawal bleeding.

Five Steps of Breast Self-Exam (BSE):

Step 1:

Stand in front of the mirror with your shoulders straight and your arms on your hips and look at your breasts (Breast awareness) and see:

- At their usual size, shape, and colour.
- That they are evenly shaped without visible distortion or swelling.
- Consult doctor if there is:
    - dimpling, puckering, or bulging of the skin
    - changed position or an inverted nipple (pushed inward instead of sticking out)
    - redness, rash, or swelling of the breasts

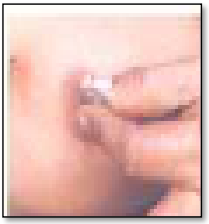


Step 2:



- Now, raise your arms and look for the same changes.
- Look for any dimpling of skin or in-drawing of nipple.

Step 3:



- While you're at the mirror, gently squeeze each nipple between your fingers and thumb.
- Consult doctor if there is persistent nipple discharge.

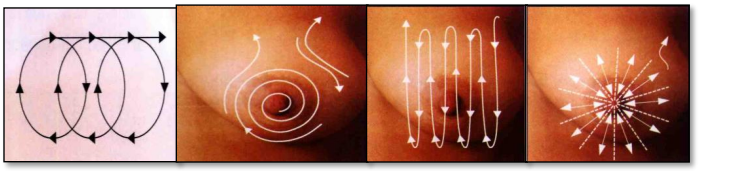
Step 4:



- Lie down and use your right hand to feel your left breast and then your left hand to feel your right breast.
- Use a firm, smooth touch with the first few fingers of your hand, keeping the fingers flat and together.
- Cover the entire breast from top to bottom, side to side—from your collarbone to the top of your abdomen, and from your armpit to your cleavage.



- Follow a pattern to be sure that you cover the whole breast. Begin at the nipple, moving in larger and larger circles until you reach the outer edge of the breast. Also move your fingers up and down vertically, in rows.
- Be sure to feel all the breast tissue: just beneath your skin with a soft touch and down deeper with a firmer touch. Begin examining each area with a very soft touch, and then increase pressure so that you can feel the deeper tissue, down to your ribcage.



Step 5:



- Feel your breasts while you are standing or sitting.
- It is easier to feel the breasts when their skin is wet and slippery (like while taking bath).
- Cover your entire breast, using the same hand movements described in Step 4.

Clinical Breast Examination

Clinical breast examinations (CBE) for women preferably in the age group 40-69 years should be encouraged. Yearly CBE can be feasible which can be practiced by General Practitioners (GPs) or trained health professionals.

Steps of Clinical Breast Examination

i. The breasts are inspected with the patient lying down to look for any asymmetry.

ii. Then with the flat of the hand, both the breasts are palpated in a circular manner starting from the nipple and areolae in a clockwise manner towards the periphery and the axillary tail of the breast in sitting and lying down position. Then the axilla, supraclavicular region and liver are also examined.

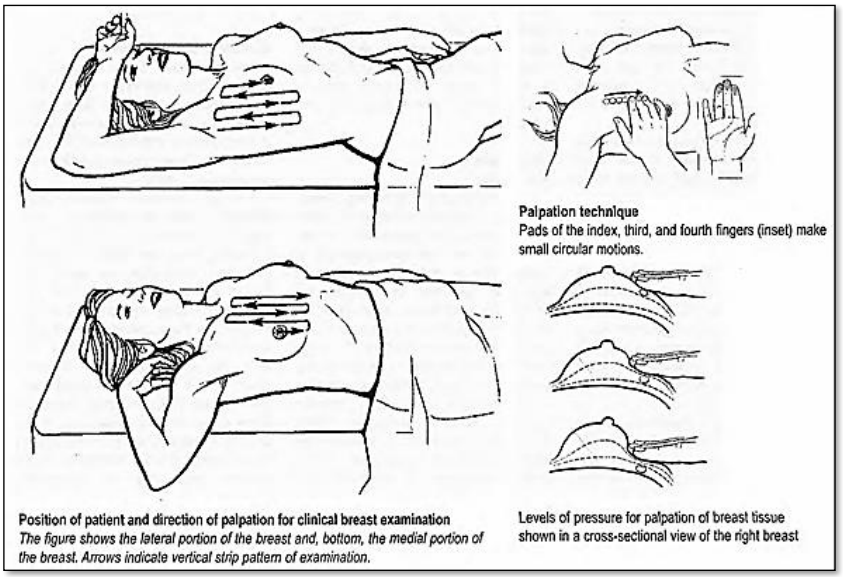
iii. Visual examination of the woman in three different standing positions; arms relaxed at her sides; hands pressed firmly on her waist and leaning forward; and arms over her head. The examiner seeks subtle asymmetries in the appearance of the breasts;

iv. palpation of the supraclavicular and axillary nodes with the woman seated and re-palpation of the axillary nodes with the woman supine;

v. vertical-strip search of the breasts over an area extending from the mid-axillary line to the mid-sternum and from above the sub-costal margin (fifth rib) to the clavicle, including palpation of the nipple and areola;

vi. Application in this search of three levels of pressure, superficial, medium and deep, at each palpation site. Palpation is done with the finger pads of the three middle fingers, and pressure is applied with circular motions at each site. For the lateral half of the breast, the torso is rotated in the medial direction; for the medial half of the breast, the torso is rotated laterally in order to spread out the breast tissue;

vii. When an abnormality is detected, the corresponding area of the other breast is examined. If the finding is not bilateral, further investigation is required.



Position of patient and direction of palpation for clinical breast examination
The figure shows the lateral portion of the breast and, bottom, the medial portion of the breast. Arrows indicate vertical strip pattern of examination.

Palpation technique
Pads of the index, third, and fourth fingers (inset) make small circular motions.

Levels of pressure for palpation of breast tissue shown in a cross-sectional view of the right breast

Clinical examination of Breast would be done by Physician/Surgeon to detect early stages of Breast cancer. Suspicious cases would be referred to higher centers for further investigations by Fine Needle Aspiration Cytology etc.

Following women have a higher than average risk of developing breast cancer, hence more attention is given to them in screening.

1. Positive Family History: One or more family member (blood relation) has a history of breast or ovarian cancer
2. Genetic Predisposition: One or more family member (blood relation) is known to harbour a 'genetic' abnormality causing breast cancer. This significantly increases the risk of developing a breast cancer
3. Previous Radiation Therapy to chest wall: If a woman has received radiation therapy to chest wall for any cause, she stands an increased risk of developing breast cancer

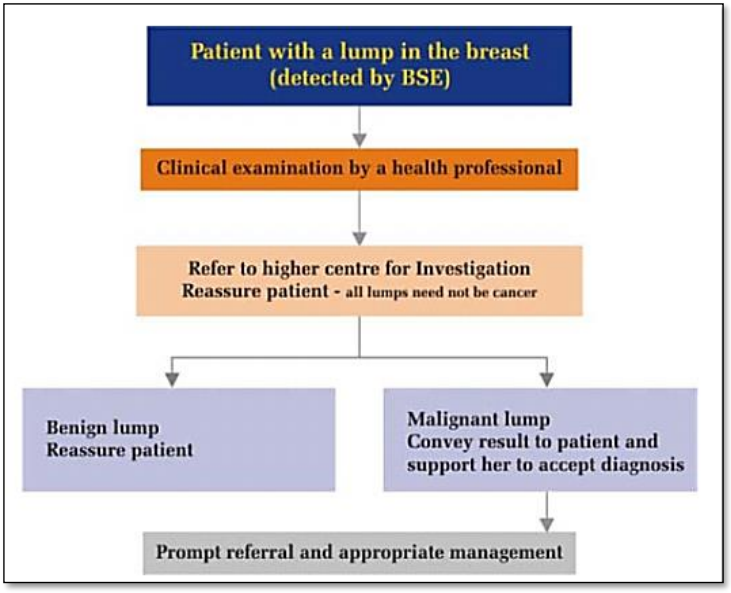


Figure 20: Flowchart of management decisions in breast cancer

Protocol for Early Detection for Oral Cancer

Introduction to Tobacco Use and Oral Cancer

Oral cancer is a major problem in the Indian subcontinent where it ranks among the top three types of cancer in the country. Incidence of oral cancer in India is high that is, 20 per 100,000 population and accounts for over 30% of all cancers in the country. Oral cancer refers to all cancers of the oral cavity, which includes the following:

- lips
- tongue
- gums (gingiva)
- lining inside the lips and cheeks (labial mucosa and buccal mucosa)
- floor of the mouth
- roof of the mouth (palate)
- the area behind the wisdom teeth (retro molar trig one)

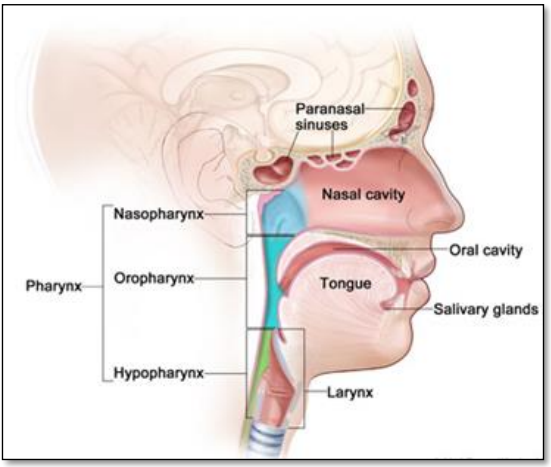
Most oral cancers are located on the sides of the buccal mucosa, lower alveolus and tongue.

Oral Cancer Symptoms

- a sore on the lip or in the mouth that does not heal, after 2 weeks
- a lump on the lip or in the mouth or throat
- a white(leukoplakia) or red(erythroplakia) patch on the buccal mucosa, palate and tongue
- unusual bleeding, pain or numbness in the mouth
- a sore throat that does not go away, or a feeling that something is caught in the throat
- difficulty or pain with chewing, swallowing, talking
- swelling of the jaw that causes dentures to fit poorly or become uncomfortable
- a change in the voice and/or pain in the ear
- swelling in neck

Oral Cavity: Anatomy

The oral cavity begins at the border between the skin and the lips (vermillion border). The roof of the mouth is formed by the hard palate. The oral cavity leads into the oropharynx, which includes the soft palate, the back of the tongue and the tonsils. The inner surface of the cheeks forms the sides of the oral cavity. The lowest part of the oral cavity is the floor of the mouth, which is covered by the tongue.



The oral cavity can be divided into specific areas, including:

- lips
- labial mucosa (inner lining of the lips)
- commissure of lips (where the upper and lower lips meet at the corner of the mouth)
- vestibule (a space bounded by the teeth and gums on the inside and the mucosal surface of the lips and cheeks on the outside)
- oral tongue (the front two-thirds of the tongue)
- floor of the mouth
- buccal mucosa (the inner lining of cheeks)
- gingiva (gums)
- retromolar trigone (the area just behind the back molars in the lower jaw)
- hard palate (the bony part at the front of the roof of the mouth)
- teeth
- lower jaw (mandible)
- upper jaw (maxilla)

Brief Description of Types of Oral Pre-Cancerous Lesions

Leukoplakia

Oral leukoplakia is now defined as “a predominantly white lesion of the oral mucosa that cannot be characterized as any other definable lesion”. In addition, a recommendation has been made to distinguish between a provisional clinical diagnosis and a definitive one. A provisional clinical diagnosis of oral leukoplakia is made when a lesion at the initial clinical examination cannot be clearly diagnosed as any other definable lesion of the oral mucosa with a white appearance. In such situations a definitive diagnosis is made by possible identification and elimination of suspected causative factors like chemical burn, frictional keratosis, habitual cheek or lip biting, papilloma, lupus erythematosus, white sponge nevus, candidiasis, lichen planus including lichenoid lesions, etc. Two to four weeks is a reasonable interval to observe the possible regression or disappearance of the white lesion after elimination of possible causative factors before subjecting the person to a biopsy. Leukoplakias are asymptomatic and are usually noticed by healthcare providers during routine examinations. It is diagnosed both by its clinical appearance and by the exclusion of other lesions that present as oral white lesions.



Figure 22: Leukoplakia

Proliferative Verrucous Leukoplakia

Proliferative verrucous leukoplakia is a rare clinical form of oral precancer, otherwise known as florid papillomatosis, characterised by multiple squamous papillary nodules. This is an aggressive form of oral leukoplakia, and nearly all cases of proliferative verrucous leukoplakia eventually undergo malignant transformation at multiple sites.

Erythroplakia

Erythroplakia is defined as a fiery red patch that cannot be characterized either clinically or pathologically as any other definable lesion. These may appear as smooth, velvety, granular or nodular lesions often with a well-defined margins adjacent to normal looking mucosa. The soft palate, the floor of mouth, the ventral surface of tongue and the retromolar area are the most common sites of involvement.

Erythroplakia is more common among middle aged to elderly persons and, especially among men. The prevalence of these lesions range from 0.02-0.83% in different regions. Erythroplakia is strongly associated with tobacco habits and alcohol drinking. The risk factors for erythroplakia are the same as for oral squamous cell carcinoma

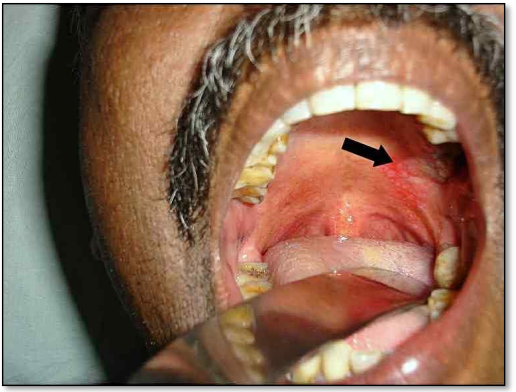


Figure 23: Erythroplakia

Oral Sub-Mucous Fibrosis (OSMF)

Oral submucous fibrosis (OSF) is a well-recognized oral precancerous condition, observed predominantly in populations of South Asian ethnic origin. Thus, it predominantly occurs in the Indian subcontinent and people of South Asian ethnicity living in other countries such as the UK, Singapore and Malaysia, among others. It is characterized by a unique generalized fibrosis of the submucosal oral soft tissues, resulting in marked rigidity of the oral mucosa leading to progressive inability to open the mouth, rigidity of lips and difficulty in protruding the tongue. The prevalence of OSF ranges from 0.2-1.2% in India



Figure 24: Oral Sub-mucous

Self-Examination of oral cavity

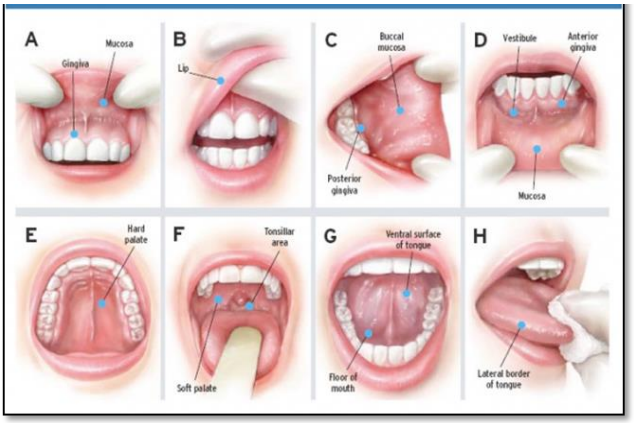
Self-Examination of oral cavity can detect oral lesions at an early stage All habitual tobacco users should do it once a month.

How to do it:

- Rinse the mouth with water and stand before a mirror in adequate light
- Look in the mirror for any abnormal white or red patch, ulcer or roughened area, granular area or swelling in the mouth
- If any such area is seen, the suspicious area should be felt with the fingers. Normal oral mucosal is soft and pink.
- Consult a doctor if you find any abnormal area.

Oral Cancer Screening by Oral Visual Inspection

Steps of performing the Oral Examination



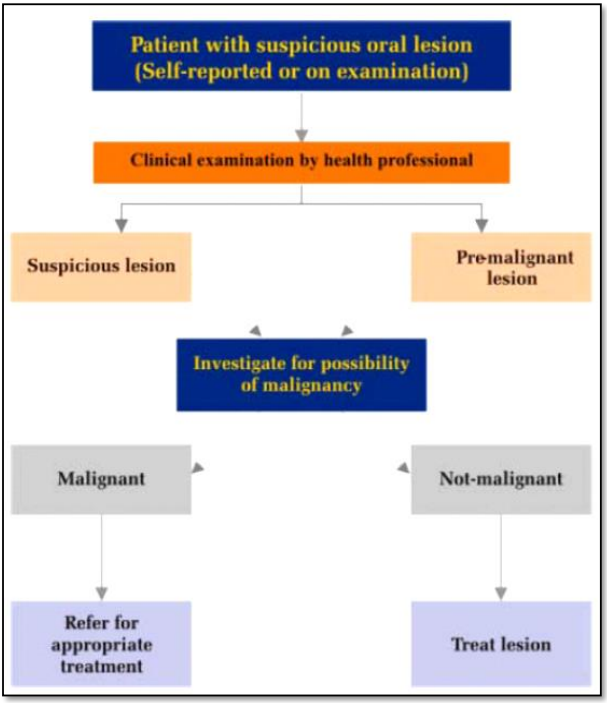


Figure 25: Flowchart of management decisions in Oral cancer

Behavioural Counselling for Tobacco Cessation

Brief advice for quitting tobacco use consists of 5 'A's

1. Ask: Ask the patient if he/she is a tobacco user, at every visit.
2. Advise: Briefly advise against continuing tobacco use and link the current condition/ailment to continued tobacco use, where possible e.g. "Quitting smoking/tobacco use would improve your health and will aid in early recovery".
3. Assess: Assess readiness to quit by asking the patient whether he or she is ready to quit at this time. e.g. "How recently have you thought about quitting tobacco?"

If the patient appears ready to change (quit), next steps are:

1. Assist: Assist the tobacco user in making a quit plan.
2. Arrange: Arrange for follow-up by setting the next contact.

Approach for a current tobacco user who has not considered quitting tobacco use (5 'R's)

If the tobacco user is not yet thinking about quitting tobacco use, the doctor will promote greater awareness of the relevance to the patient of the advice to quit, the risks of use and rewards (benefits) of quitting. Many tobacco users are largely unaware of the potential harm that tobacco use can do to them. If the patient is not ready to quit, the doctor must not push the patient. People usually need time to change (incremental nature of change).

If the patient is at least thinking about quitting, the doctor can find out the patients' roadblocks to quitting and help the patient see ways to overcome these. This process may be enough to help the patient get ready to quit (without pushing). At the next visit, this process should be repeated so that the information about relevance, risks of continuing and rewards of quitting can sink in a little more and some roadblocks can be removed.

As you can see, the doctor must try to make the tobacco user think about quitting. This is important because there are so many other forces acting that are difficult to control, physiological compulsions to use tobacco, learned habits, social pressures, accessibility etc. Engaging the mind of the tobacco user, bolstering it with new knowledge and a sense of caring by the person counseling can help motivate him/her to change. Follow-up is important to help keep the tobacco user on track until he or she is confident about remaining tobacco free.

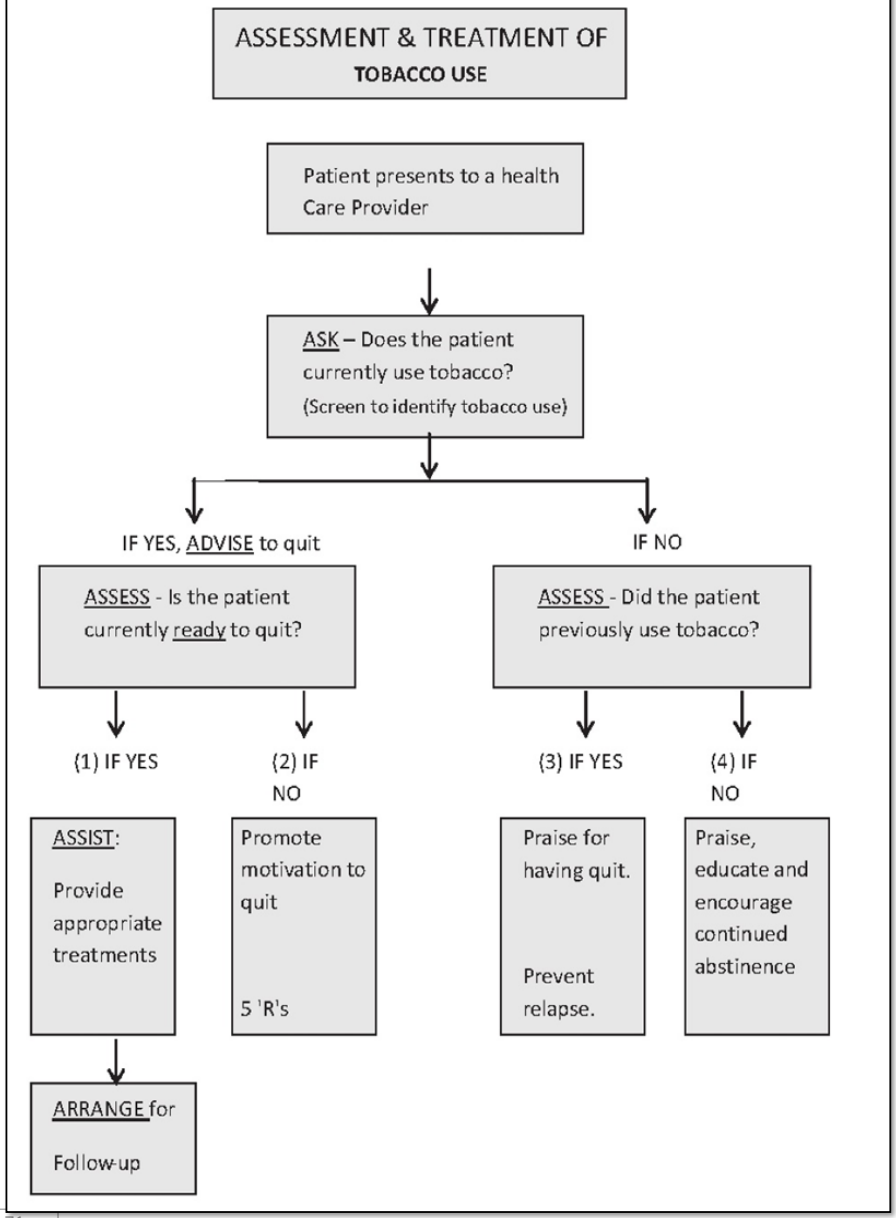
The 5'R's of behavioral counselling are:

1. Relevance: Explain the relevance of quitting to the client and harmful effects of tobacco use.
2. Risks: Highlight the health hazards that are more relevant to the individual tobacco user.
3. Rewards: Benefits of quitting all forms of tobacco use should be explained to the tobacco user (Health, financial, approval of family etc.)
4. Roadblocks: Barriers that the client may face in his/her quit attempt should be identified. Withdrawal symptoms, fear and concern associated with quitting, depression, lack of social support, enjoyment of tobacco are some of the barriers that the client may face in an attempt.
5. Repetition: The physician should assure the client that because of chronic nature of tobacco dependence, relapses are common in the initial phases and multiple attempts may have to be made before he/she is able to quit tobacco.

Health Professional can be trained to examine oral cavity especially of tobacco users. All parts of the oral cavity should be examined; oral cavity includes lip, anterior 2/3 of tongue, floor of mouth, buccal mucosa, gingival mucosa, hard palate and retromolar trigone.

Any patients with suspicious lesion should be referred for biopsy and further management at higher centers.

Intervention Method Algorithm for Quitting Tobacco Use



Principles of Pharmacotherapy in Tobacco Cessation

Some drugs have been studied for their potential to help smokers and tobacco users quit smoking/tobacco use. Research is ongoing to develop a wider range of such medications and broaden the effects. Pharmacotherapy, is the use of medications to minimize the effects of craving and other withdrawal symptoms and make quitting easier. The rationale goes that if a patient is told that he can be given a medication to reduce withdrawal symptoms, which he fears, he may be more motivated to try quitting (decisional balance). Such drugs include pure nicotine, as a gum, lozenge, nasal spray or inhaler or patch, antidepressants or other medications as oral pills.

The fist type of pharmacotherapy developed for smoking cessation was nicotine replacement therapy (NRT), forms of purified nicotine, which are still being prescribed. Next, antidepressants were found effective for smoking cessation: bupropion (first choice), and nortriptyline (second choice) (Hughes et al, 2006). Varenicline (Nicotinic acetylcholine receptor partial agonist) is the latest addition and have shown higher quit rates in the clinical trials. Thiswas approved by FDA in 2006 and was introduced in India in 2008.

Informed consent for Pharmacotherapy

Be sure to explain to your patient just what the expected effect of the prescribed medicine is and what are the possible side-effects.

When to Prescribe Pharmacotherapy

All persons with severe dependence. A score of more than 6 on the Fagerstrom questionnaire

- Tobacco users with multiple failed self-attempts.
- Tobacco users unable to abstain with brief intervention alone.

The following clinical indicators may be used to assess dependence.

1. Using tobacco within the first 30 minutes of waking up.
2. Using more than 20 cigarettes or bi dis every day or using very high quantity of smokeless tobacco.

Pharmacotherapy for smokeless tobacco users

The nicotine gum has been used for smokeless tobacco users. High-dose nicotine gum and patch therapy is safe and well tolerated by heavy smokeless tobacco users and reduces withdrawal symptoms.

Many studies have found buproprion (an antidepressant) effective for cessation of smokeless tobacco use. The most important interventions for smokeless tobacco users are an oral examination and counselling (behavioural intervention).

Method of using nicotine gum

Nicotine gum is available in India without a prescription (2 mg.), however medical supervision is important as there are precautions and to help patients achieve success in quitting. Medical prescription is required for Nicotine gum 4 mg.

How to prescribe Pharmacotherapy

	Dosage and duration	Side effects	Contraindications[2]
1. Nicotine Replacement Therapy (NRT)[1]			
a. Nicotine gum	For 1-24 cigarettes/ bidis - 2mg gum (up to 24 pieces/ day) for 12 weeks For ≥25 cigarettes/ bidis - 4mg gum (up to 24 pieces/ day) for 12 weeks Chewers need about half or a quarter of the dose as prescribed for smokers.	Mouth soreness, burning in the mouth, throat irritation, dyspepsia, nausea, vomiting, hiccups and excess salivation	Gastric Ulcers, myocardial infarction or stroke in the past two weeks or poorly controlled cardiovascular disease.[3] If a patient has any serious medical condition, refer to an appropriate specialist.
b. Nicotine patch	21mg/24 hours for 4 weeks then 15mg/24 hours for 2 weeks followed by 7mg/24 hours for 2 weeks.	Local skin reaction, insomnia	Myocardial infarction or stroke in the past two weeks or poorly controlled cardiovascular disease[3] If a patient has any serious medical condition, refer to an appropriate specialist.
c. Nicotine inhaler	6-16 cartridges/day for 6 months	Local irritation of mouth and throat	- DO -
d. Nicotine nasal spray	1-2 doses/hour for 3 to 6 months	Nasal irritation, irritation of throat, coughing and watering of eyes.	- DO -
2. Non-NRT			
a. Bupropion	150mg OD for 3days followed by 150mg BD for 7 to 12 weeks. Bupropion is started while the person is still using tobacco. The person can completely quit two weeks after initiating bupropion.	Agitation, restlessness, insomnia, gastrointestinal upset, anorexia, weight loss, headache and lowering of seizure threshold (at doses above 600 mg/day). Rarely allergic reactions can occur, including skin rashes, fever, muscle and joint pain.	History of allergy, tumours of central nervous system, severe liver diseases, undergoing unsupervised withdrawal of alcohol or benzodiazepenes, uncontrolled seizures, pregnant and lactating women, those below 18 years, and persons on monoamine oxidase inhibitors.

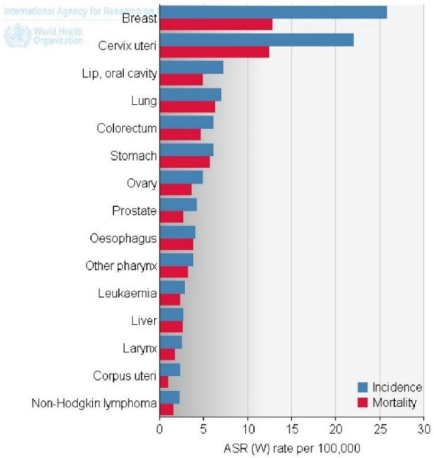
	Dosage and duration	Side effects	Contraindications[2]
b. Varenicline	Initially 0.5 mg once daily for the first three days, increased to 0.5 mg twice daily for the next four days, and then increased to 1mg twice daily for 12 weeks.The person can quit one week after initiating Varenicline	Agitation, depression, restlessness, insomnia, bad dreams, suicidal ideations, gastrointestinal upset and headaches. Allergic reactions may occur rarely.	Pregnant women, children or people with mental illness.Stop treatment if changes in mood & behavior, agitation and suicidal ideations occur.

Principles of Cancer Management: A Short Overview

Introduction

Cancer is a chronic disease like any other chronic medical condition. Hence, when a patient presents to us in early stages, our aim is to cure the cancer, and in very advanced cases or when the disease has already spread, our main aim is to control the symptoms to improve quality of life and prolong the patient`s life span as much as possible.

Estimated age-standardised incidence and mortality rates: both sexes



Estimated incidence, mortality and 5-year prevalence: both sexes

Cancer	Incidence			Mortality			5-year prevalence		
	Number	(%)	ASR (W)	Number	(%)	ASR (W)	Number	(%)	Prop.
Lip, oral cavity	77003	7.6	7.2	52067	7.6	4.9	118902	6.6	13.5
Nasopharynx	3947	0.4	0.3	2836	0.4	0.3	9967	0.6	1.1
Other pharynx	38691	3.8	3.7	32784	4.8	3.1	56754	3.2	6.4
Oesophagus	41774	4.1	4.1	38683	5.7	3.8	22157	1.2	2.5
Stomach	63097	6.2	6.1	59041	8.6	5.7	45390	2.5	5.1
Colorectum	64332	6.3	6.1	48603	7.1	4.6	86650	4.8	9.8
Liver	27416	2.7	2.7	26763	3.9	2.6	13089	0.7	1.5
Gallbladder	18787	1.9	1.8	15866	2.3	1.5	22892	1.3	2.6
Pancreas	11936	1.2	1.2	10828	1.6	1.1	6730	0.4	0.8
Larynx	25446	2.5	2.5	17560	2.6	1.7	50494	2.8	5.7
Lung	70275	6.9	6.9	63759	9.3	6.3	32464	1.8	3.7
Melanoma of skin	2103	0.2	0.2	1122	0.2	0.1	5314	0.3	0.6
Kaposi sarcoma	19	0.0	0.0	14	0.0	0.0	26	0.0	0.0
Breast	144937	14.3	25.8	70218	10.3	12.7	396991	22.2	92.6
Cervix uteri	122844	12.1	22.0	67477	9.9	12.4	308901	17.3	72.0
Corpus uteri	12325	1.2	2.3	4773	0.7	0.9	44980	2.5	10.5
Ovary	26834	2.6	4.9	19549	2.9	3.6	55231	3.1	12.9
Prostate	19095	1.9	4.2	12231	1.8	2.7	63818	3.6	14.1
Testis	3298	0.3	0.5	1452	0.2	0.3	11079	0.6	2.4
Kidney	9658	1.0	0.9	5973	0.9	0.6	19977	1.1	2.3
Bladder	16273	1.6	1.6	9523	1.4	0.9	36904	2.1	4.2
Brain, nervous system	18831	1.9	1.6	15152	2.2	1.4	27408	1.5	3.1
Thyroid	13904	1.4	1.2	3290	0.5	0.3	55539	3.1	6.3
Hodgkin lymphoma	8371	0.8	0.7	4342	0.6	0.4	13629	0.8	1.5
Non-Hodgkin lymphoma	23801	2.3	2.2	16597	2.4	1.5	27099	1.5	3.1
Multiple myeloma	6955	0.7	0.7	6027	0.9	0.6	11886	0.7	1.4
Leukaemia	32532	3.2	2.8	26712	3.9	2.3	24042	1.3	2.7
All cancers excl. non-melanoma skin cancer	1014934	100.0	94.0	682830	100.0	64.5	1790498	100.0	202.9

http://globocan.iarc.fr/Pages/fact_sheets_population.aspx

Approach to Cancer Management

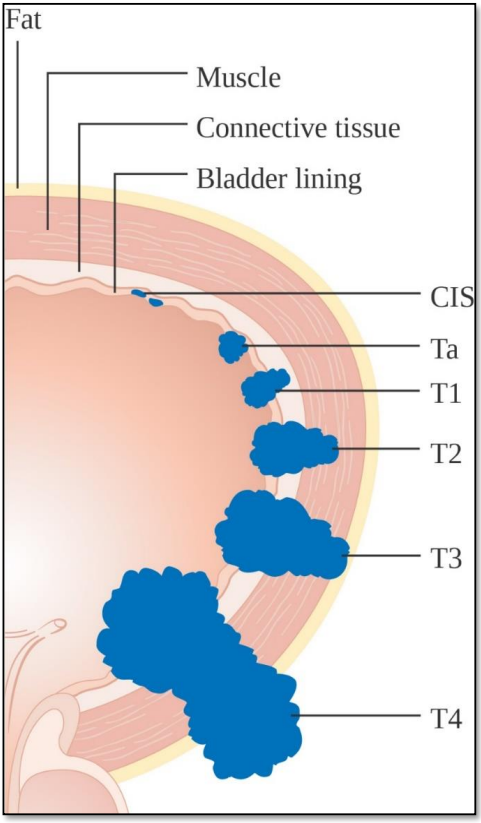
Three main questions to consider are:

1. **What is the type of cancer?**
   Cancers are broadly classified into two types haematological (blood cancers) e.g. lymphoma, leukaemia's and solid cancers eg oral cavity, breast, cervix, etc. Establishing diagnosis requires a tissue diagnosis. Tissue diagnosis may be done by FNAC (aspiration of cells by fine needle) or biopsy (removal of bits of tissue by large bore needle from tumour or open biopsies). In modern oncology, without tissue diagnosis, it is unusual or inappropriate to start treatment based on clinical diagnosis alone. Tissue diagnosis is also important to perform molecular studies to select appropriate medical therapies.
2. **What is the extent of the spread of the cancer?**
   The extent is assessed by one or more of the following imaging, which includes X-rays, Ultrasonography, CT scans, MRI, bone scan and PET scans.
3. **Is it curable or not curable?**
   This depends on the type of cancer, its extent and nodal spread. Also presence or absence of and also the extent of distant metastasis.

   A simple staging system for all types of cancer groups, which helps health care providers across all fields to understand the stage of the cancer and also utilize the same to explain patients is described below.

- **In situ**—abnormal cells are present but have not spread to nearby tissue.
- **Localized**—Cancer is limited to the place where it started, with no sign that it has spread.
- **Regional**—Cancer has spread to nearby lymph nodes, tissues, or organs.
- **Distant**—Cancer has spread to distant parts of the body.

The TNM system is the most widely used scientific cancer staging system, used by doctors is described in detail later.



- **T**: size or direct extent of the primary tumour
  - Tx: tumour cannot be evaluated
  - Tis: carcinoma in situ
  - T0: no signs of tumour
  - T1, T2, T3, T4: size and/or extension of the primary tumour

- **N**: degree of spread to regional lymph nodes
  - Nx: lymph nodes cannot be evaluated
  - N0: tumour cells absent from regional lymph nodes
  - N1: regional lymph node metastasis present; (at some sites: tumour spread to closest or small number of regional lymph nodes)
  - N2: tumour spread to an extent between N1 and N3 (N2 is not used at all sites)
  - N3: tumour spread to more distant or numerous regional lymph nodes (N3 is not used at all sites)

- **M**: presence of distant metastasis
  - M0: no distant metastasis
  - M1: metastasis to distant organs (beyond regional lymph nodes

Once diagnosis and staging is achieved, a treatment plan is formulated. Increasingly, cancer is managed in a multidisciplinary team setting to improve outcome and decrease the morbidity of treatment. Multidisciplinary or team care brings together a group of health professionals with appropriate skills to consider treatment and care options. They develop the best treatment plan together, rather than each specialist seeing the patient individually to give their opinion. Members of multidisciplinary team include surgeons, radiation and medical oncologists/haematologists, radiologists, pathologists, general practitioners, nurses and allied health professionals and palliative care physicians. Cancer care coordinators also play an important role in the provision of coordinated care.

Roles of various multidisciplinary team members

Specialties	Examples of roles
Medical oncologist	Chemotherapy, biological therapy, hormonal therapy, immunotherapy
Oncological surgeons	Definitive and palliative surgery
Radiation oncologist	External beam radiotherapy, brachytherapy, nuclear medicine.
Allied health: Dietician, physiotherapy, occupational therapy, social work, pharmacy and psychology	Relevant to each specialty
Cancer nurses	Care coordination, clinical trials, chemotherapy administration
Palliative care	Symptom control, hospice care

A brief description of the services provided by the members of multi-disciplinary team is given below. The three main modalities of treatment are surgery, chemotherapy and radiation. In select cases like early oral cavity cancers and cervical cancer only a single modality of treatment like surgery or radiation will suffice. But in vast majority of cases a combination of more than one modality is required to give adequate cure.

Surgery

Surgery is the most important integral part in management of all solid tumours (e.g. breast, oral cavity cancers etc.) in curative settings, it involves removal of primary cancer and regional lymph nodes. Surgery can be done upfront in early stages of cancer. In advanced cases, chemotherapy can be given to downstage the disease (neo-adjuvant) and then taken for definitive surgery. In locally advanced and metastatic patients where definitive treatment is not possible, palliative procedures to reduce the symptoms can be done.eg. Intestinal bypass in obstructed bowel cancers, toilet mastectomies in fungating tumour of breast. In very select cases, where the disease has limited spread to liver or lung, surgical removal of these lesions is possible (metastatectomy). Also, reconstructive surgeries post major resection also plays a very vital role in recent times e.g. whole breast reconstruction following removal of breast. Curative cancer surgeries are done by surgical oncologists or by general surgeons adequately trained in cancer surgeries.







Figure 30: Palliative Toilet Mastectomy: Pre-Op and Post-Op

Medical therapies including chemotherapy, biological therapy, hormone therapy and immunotherapy

Medical therapies and radiotherapy are the only line of definitive management for all types of hematological cancers. Medical therapies forms an integral part in treatment of all solid cancers. The various drugs are explained in detail in later sections. Chemotherapy is given under various settings as described below.

Curative

E.g. Leukaemia, lymphoma, germ cell tumours, choriocarcinoma, Ewing sarcoma/osteosarcoma, etc.

Neo adjuvant

To reduce the disease burden (downstage) before definitive surgery or RT, in combination with RT or alone

Eg breast, oesophagus, stomach, etc.

Adjuvant

to eliminate micro metastatic disease after surgery or radiotherapy

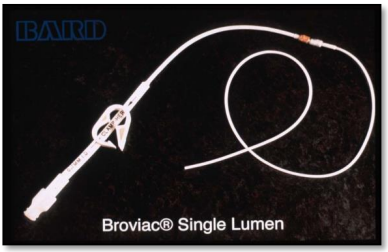
E.g. Breast, colon, ovarian, sarcoma, etc.

Concurrent with radiation therapy (as radio sensitizers)

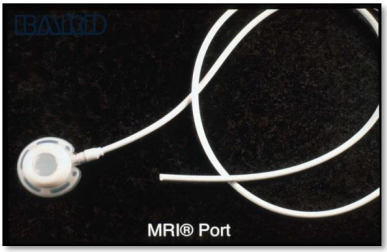
E.g. Head and neck cancer, cervical cancer, etc.

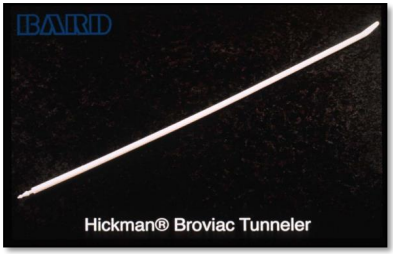
Palliative (to improve quality of life and prolong survival)

E.g. All Metastatic cancers

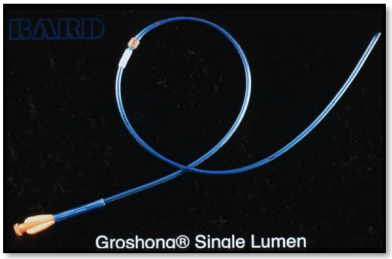

BAIRD
Broviac® Single Lumen


BAIRD
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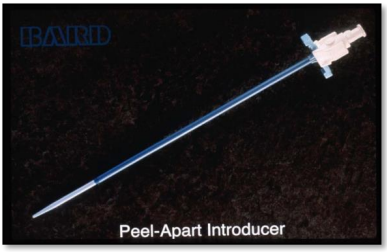

BAIRD
Hickman® Broviac Tunneler


BAIRD
MRI® Port with Groshong®


BAIRD
Groshong® Single Lumen


BAIRD
Winged Infusion Sets


BAIRD
Groshong® Dual Lumen (pre-insertion)


BAIRD
Peel-Apart Introducer

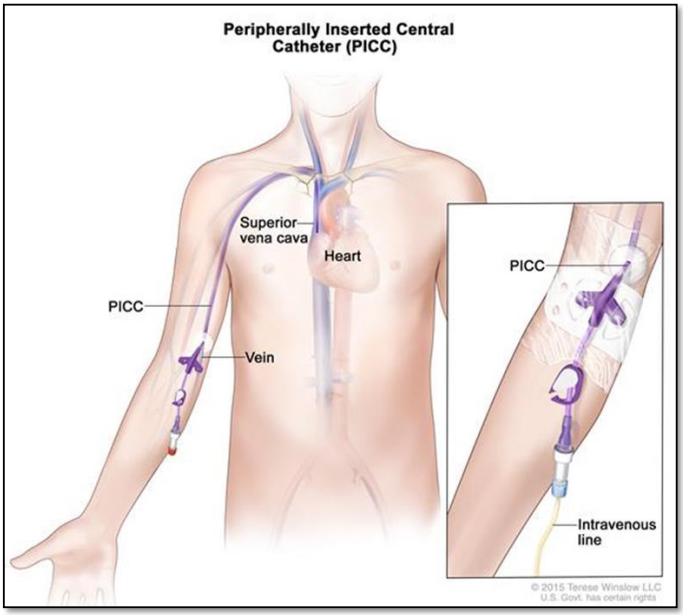




Tunneled Catheter



Child with PICC



Radiation Therapy



Radiation is given 2 ways: either high-energy rays are aimed from a machine (external radiation) or implants (radiation sources) are put into the body (placed) near the tumour.

External radiation: Getting external radiation is painless, much like having an x-ray taken. It's usually done in an outpatient setting, and the treatments take very little time. Treatment is most often given 5 days a week for 5 to 8 weeks, depending on the size, place, and type of cancer being treated.

Radiation implants: In some cases, radiation may be given through implants placed (temporarily or permanently) inside the body (Brachytherapy).

Like chemotherapy, radiation therapy is also given under various settings.



Curative

E.g. Head and neck, cervical, lung, prostate,etc.

Adjuvant (to improve local control post-surgery)

E.g. head and neck, breast, brain,etc.

Neo adjuvant

To reduce the disease burden (downstage) before definitive surgery, in combination with chemo therapy or alone.

Oesophagus, rectum, soft tissue sarcomas,etc.

Palliative (to improve symptoms)

E.g. In Advanced local disease and metastatic cancers, to reduce pain in bone metastasis.

Palliative Care

Aim of palliative care is to improve quality of life by controlling symptoms

If a tumour can be shrunk by chemotherapy or radiotherapy, this would be an efficient option for controlling symptoms. In most cases, concurrent use of palliative care services and active anti-cancer therapy are necessary to maintain quality of life. In all these situations, it is important to identify and treat the cause of the symptoms**.**

Examples of symptoms and management are outlined below:

Pain: simple analgesics, narcotics, parenteral narcotics, etc.

Nausea: metoclopramide, 5HT3 antagonist, steroids, etc.

Loss of appetite: steroids, megestrol, etc.

Cough/SOB: codeine, narcotics, nebulizers, etc.

Depression: control symptoms, correct causes, counselling, antidepressants

family support, aids, home visits, physiotherapy, nutrition, occupational therapy, social work

Of all the above examples pain is the most underestimated condition in terminally ill cancer patients. Hence all health care providers are requested to put their best efforts for pain relief of these patients.

Rehabilitation therapist, such as physical, occupational, speech, or recreational therapists. These professionals help people with cancer return to their highest level of independence. For example, they can help people with oral cavity cancer regain speech and swallowing after surgery, or, help women with breast cancer learn exercises to regain strength after a mastectomy.

Dietician ensures proper nutrition to all cases patients.



Cancer Management in Detail

Evaluation

A thorough physical examination is a must for all cancer patients. Proper clinical examination of primary site, regional lymph nodes and metastatic sites gives an idea about the disease extent. It helps in getting tissue biopsies from appropriate sites, ordering necessary imaging investigations and to some extent prognosticate patient e.g. palpable supraclavicular node in gastrointestinal (GI) malignancies has a very grim prognosis. Also assessment of performance status is a must, as it gives a clear picture, whether the patient can tolerate radical surgeries or toxic chemotherapy or radiation schedules. Eliciting family history is vital, especially in young patients as it helps in identifying cancer at early stage in other family members. Following this, routine blood investigations and serum tumour markers depending on the site of cancer are ordered.

Imaging

All solid tumours require adequate and site-specific imaging. This facilitates diagnosis and staging cancer. The goals of imaging the primary tumour are to assess tumour size, invasion into surrounding structures, nodal involvement and distant metastasis. The various imaging modalities are X-rays, ultrasound (USG), CT scans, MRI, mammogram, nuclear imaging like PET and bone scan. One or more of the above are used to assess the extent of the disease. Though PET scan is used to image patients, it is not mandatory for all. Also in cases of GI malignancies endoscopy or colonoscopy is done to assess site and to take biopsies of tumour.

Pathological biopsy

Despite suggestive imaging, a cancer is diagnosed only after appropriate cytological (FNAC) or histopathological examination. FNAC can be done to assess disease in a node or USG guided for liver lesions to prove metastasis in a known cancer patient. Biopsies may be undertaken percutaneously -- for example, a core biopsy of the breast,

with an endoscope in stomach and colorectal cancers or by surgery (excision or incision biopsies).

A biopsy should confirm the tumour type, grade, may show lymphovascular invasion and in some cases, special immunohistochemical(IHC) stains may be performed to assess hormone receptor status such as in breast cancer or flow cytometry may be performed to assess subtypes such as in lymphoma.

Staging

The TNM staging system (American Joint Commission on Cancer AJCC) is devised for cancers to allow an assessment of T- tumour, N- nodal metastases and M- distal metastases. The goal of having a site-specific staging system is to estimate prognosis, facilitate treatment planning including the sequence of treatments and allow comparisons of treatment for different stages. Generally, a combination of different 'T', 'N', and 'M' allows the cancer to be grouped into stages. Stages I-IV usually depict a tumour in the following state:

Stage 1- early and superficial cancer,

Stage 2- locally advanced,

Stage 3- regionally advanced with lymph node metastases and

Stage 4- distant metastatic disease.

Performance Status of Patient

ECOG Performance Status

Developed by the Eastern Cooperative Oncology Group

GRADE	ECOG PERFORMANCE STATUS
0	Fully active, able to carry on all pre-disease performance without restriction
1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, e.g., light house work, office work
2	Ambulatory and capable of all self-care but unable to carry out any work activities; up and about more than 50% of waking hours
3	Capable of only limited self-care; confined to bed or chair more than 50% of waking hours
4	Completely disabled; cannot carry on any self-care; totally confined to bed or chair
5	Dead

Patients with ECOG status 0, 1 and 2 can withstand radical surgery or definitive chemo radiation schedules.

ECOG -3 patients can be considered for limited palliative surgeries or palliative chemo/RT regimens. ECOG -4 receives best supportive care (BSC)

Principles of Cancer Surgery

Introduction

Cancer surgery has evolved over the decades from a radical 'one size fits all' approach to a patient-specific, cancer-specific direction, which means that surgeons rely on their multidisciplinary partners in the assessment of patients. As surgeons are frequently the first specialists involved with most solid tumours, familiarity with pre-operative imaging, pathological biopsy and patient-selection, staging and oncological surgical principles are fundamental to the surgeon's armamentarium.

Patient-selection and timing of surgery

One of the biggest challenges for the surgeon is to choose the correct surgery for the correct patient and with the tumour type and biology in mind. Although surgery removes a tumour and provides further pathological information to estimate prognosis and influence adjuvant therapies, the surgery cannot cause more morbidity than the cancer and must achieve surgical goals without compromising tumour biology.

When tumours are locally advanced, a neoadjuvant approach with chemotherapy, radiotherapy or targeted therapies may be important to 'control' the growth of a tumour, down-stage a tumour to render it operable. Similarly, patients with metastatic disease may still require surgery to prevent complications of the primary tumour, such as bowel obstruction from a colon cancer.

The pre-operative multidisciplinary team including anaesthetists, cardiologists, dieticians, physiotherapists, psychologists and tumour-specific specialist nurses help in optimizing the patient for surgery and also ensure quick post-operative recovery.

Surgery of the primary tumour

The aims of any cancer surgery are to remove the tumour with an adequate margin of normal tissue with minimal morbidity. Clear margins have an impact on local control. The concept of radicality emphasis on removal of tumour with good margin and not the organ in entirety. For e.g. tumour up to 4 cm can be removed by conserving

breast, thus avoiding mastectomy for all cases. Margin requirements differ according to the origin of the tumour and the resulting functional and cosmetic impact must be considered. For example, in order to achieve adequate margins in oral cavity cancers patient develops difficulty in speech or swallowing due to removal structures, whereas in breast cancer patients change in size, shape or even removal of breast occurs. Surgeons trained in reconstructive procedures are an important part of the team dealing with oral cavity and breast malignancies. Free flaps like thigh flaps helps in achieving good cover of huge oral cavity defects, fibula bone flaps help in improving both cosmetic and functional outcomes after removal of mandible.

Various oncoplasty procedures following breast conservation surgeries (BCS) and total reconstruction of breast using abdomen tissue are great boon to women suffering from breast cancer.

Surgery of the lymph node basin

Many solid tumours require removal of the draining lymph nodes for the purpose of staging and/or to achieve local control. When regional nodes are removed prophylactically, even if they are not involved by tumour is called elective node dissection. Elective lymph nodes dissection vary according to tumour type and may increase surgical morbidity. To avoid this sentinel node biopsy is used is some cancers, mostly in breast and melanoma. By this technique, during surgery nodes are assessed for metastasis and only if they are involved they are removed, hence avoiding prophylactic removal and its associated morbidity.



Illustration of head and neck surgery – removal of disease in cheek with neck nodes

Surgery in metastatic disease

Local control may become an issue in some patients with metastatic disease addition, in some tumour types, such as breast cancer, colorectal cancers patients having limited metastasis (oligometastasis) to lung or liver can be removed surgically (metastatectomy) along with primary surgery or at a later time when the disease recurs in these areas. It has been shown that such procedure may improve prognosis and survival. Eg. fungating breast cancer.



Pre – op



Post-op

Surgery in palliative settings

Surgery may be undertaken in an elective or emergency setting in colorectal cancer to prevent or manage a bowel obstruction or do a bypass to drain bile in pancreatic or periampullary tumours. Sometimes fungating and foul smelling tumours of breast and extremities are resected. More recently stents are deployed in obstructed oesophagus, stomach or common bile duct to palliate the patient and improve him symptomatically, thus avoiding morbidity of surgery.

Surgery for prophylaxis

Cancer surgery includes managing patients at high risk of cancer in their lifetimes, usually due to an inherited mutation such as BRCA 1 or 2 or Lynch Syndrome. Although some medications, such as tamoxifen, may reduce risk of cancer, surgery in some organs reduces the risk of cancer in that site by about 95-97%. Organs removed in surgical prophylaxis include the breasts, ovaries, colon, and thyroid.



Photo of patient on RT table with planning

Principles of Radiotherapy

Radiotherapy and multidisciplinary cancer care

Radiotherapy can prolong survival, contribute to the preservation of organs affected by malignancy, provide palliation and improve patients' quality of life. Recent research shows that about 50% of patients with newly-diagnosed cancers (other than non-melanotic skin cancers) would benefit from radiotherapy. One-quarter of these patients may need further treatment after relapse.

The provision of a safe and effective radiation oncology service is complex. It requires a substantial capital investment in radiotherapy equipment and specially-designed buildings, an ongoing investment in maintenance and replacement of the equipment, expert teams of doctors, therapists and physicists, and good access to engineering support.

How radiotherapy works

- Radiotherapy ionises chemicals within cells. The crucial lesion is DNA strand breakage which may be repaired or fixed and lead to apoptotic or mitotic cell death.
- The principle of radiotherapy is to give maximum dose to the tumour tissues while sparing the normal tissues from the radiation dose.

Mechanism of Action

Radiation therapy uses particles moving at a high frequency to target the DNA of cells in the body and change the way they are able to replicate. If the DNA required for mitosis and replication is damaged, the cells are unable replicate as usual and the growth of a cancerous tumour is inhibited.

This type of therapy manipulates the fact that cancerous cells usually replicate more quickly than normal cells in the body. As the DNA can

only be damaged when the cells are undertaking the replication process – not in the G0 phase – the majority of normal cells that take longer to replicate are less affected than the cancerous cells.

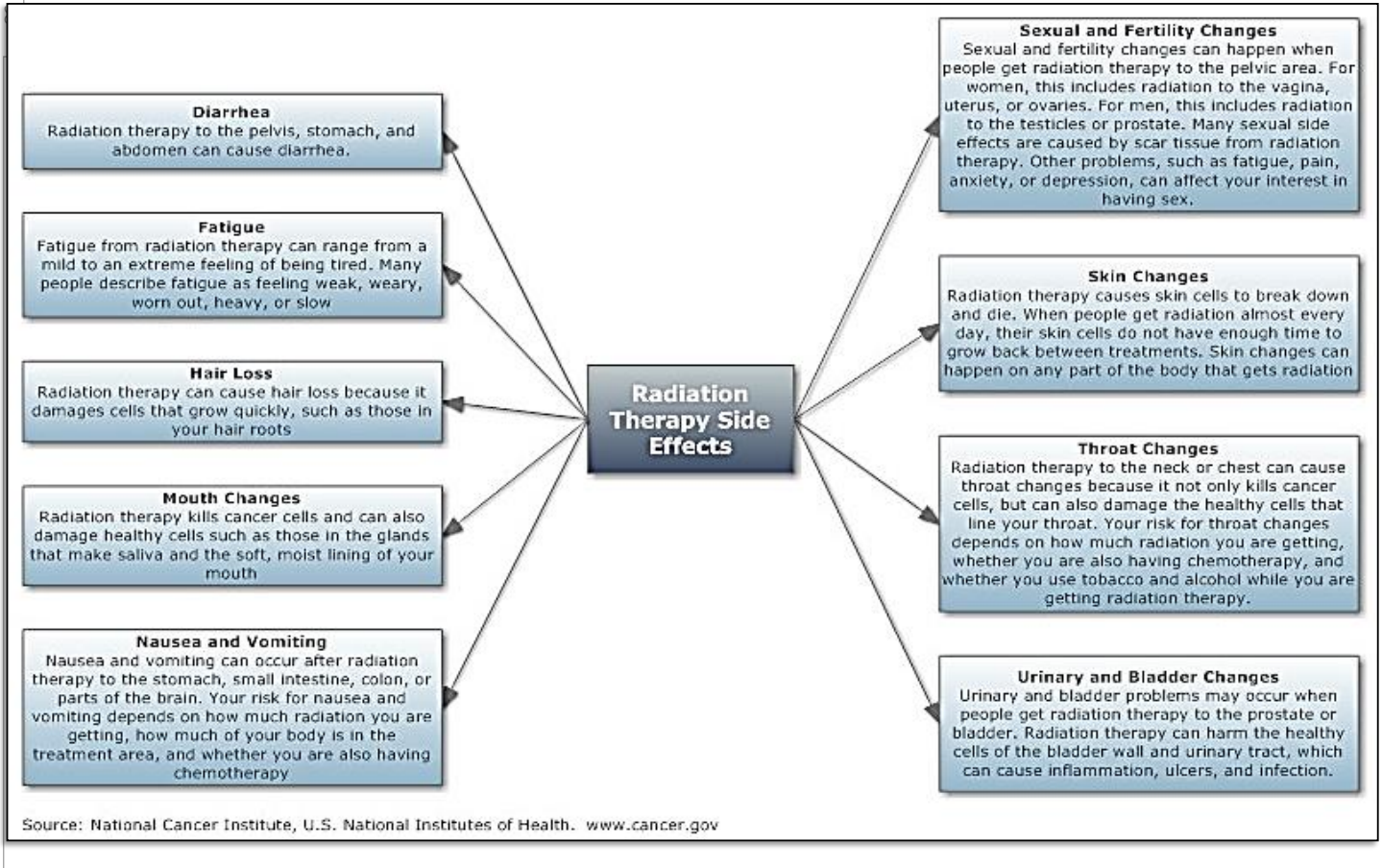
- Different types of normal and malignant cells vary in their susceptibility to ionising radiation. Clinical radiotherapy schedules are designed to exploit the differences between normal tissues and tumours, so that as many malignant cells as possible are killed, while damage to normal tissue is minimized. In radical curative treatments, total radiation doses are given in numerous small doses to maximize the damage to tumour tissues. In palliative treatments, usually large dose is given in less number of times.
- Some tumours, such as seminoma of the testis and lymphoma, are very sensitive to radiotherapy and can be treated with relatively low doses, with an expectation of cure. Other tumours, such as melanoma of the skin and glioblastoma multiforme in the brain, are notoriously resistant, even to large doses.
- A course of radiotherapy may be spread over days or weeks. This is known as fractionating, and the radiation delivered to a patient in a single treatment session is called a fraction. Fractionating allows normal tissues to repair much of the radiation damage, while tumour cells, which are less efficient at repair, do not recover. A beam of radiation is called a field. A fraction consists of one or more fields delivered sequentially.
- In general, most modern radiotherapy is delivered with very high-energy, highly-focused beams as fields which can reach deeper tumour tissues and the fields from different beams converge to deposit high dose at the tumour while depositing relatively small doses in the normal tissues through which they pass. External-beam radiotherapy can be delivered by cobalt machines or linear accelerators, collectively known as megavoltage machines.
- However, both acute (early) and late (chronic) side effects do occur. The side effects depend on several factors, including the body site being treated, the volume of normal tissue irradiated

(the larger the volume, the higher the risk and severity of side effects), the total dose, and the rate of dose accumulation (the amount per week).

- Early side effects result from damage to proliferating tissues, such as the mucosa (lining) of the gastro-intestinal tract or the skin. For example, radiotherapy to an abdominal tumour may damage the mucosa of the small bowel, causing malabsorption and diarrhoea. Most patients recover completely.
- Late reactions occur at least three months after a treatment course has ended, and are usually permanent or progressive. They usually result from damage to non-proliferating differentiated tissues, which cannot compensate for cell death by dividing to replace lost cells, for eg. Neural cells, lens of the eye. Once late effects occur, it is very difficult to reverse them, but they are very uncommon.

- Late effects, such as second cancers, may occur many years after treatment. Even relatively low doses of radiation may increase the risk of developing a malignancy. There is a long latent period after exposure. Leukaemia may appear up to 7 years after exposure and solid tumours may develop 10 or 20 years later.

- Side effects of radiation can be minimized by meticulous planning and delivery of a course of radiotherapy. Late-reacting tissues are particularly sensitive to the size of each radiation dose, so they can be protected to a large extent by giving small fractions of radiation, provided the total dose is not too high.


Radiation Therapy Side Effects
Diarrhea
Radiation therapy to the pelvis, stomach, and abdomen can cause diarrhea.
Fatigue
Fatigue from radiation therapy can range from a mild to an extreme feeling of being tired. Many people describe fatigue as feeling weak, weary, worn out, heavy, or slow
Hair Loss
Radiation therapy can cause hair loss because it damages cells that grow quickly, such as those in your hair roots
Mouth Changes
Radiation therapy kills cancer cells and can also damage healthy cells such as those in the glands that make saliva and the soft, moist lining of your mouth
Nausea and Vomiting
Nausea and vomiting can occur after radiation therapy to the stomach, small intestine, colon, or parts of the brain. Your risk for nausea and vomiting depends on how much radiation you are getting, how much of your body is in the treatment area, and whether you are also having chemotherapy
Sexual and Fertility Changes
Sexual and fertility changes can happen when people get radiation therapy to the pelvic area. For women, this includes radiation to the vagina, uterus, or ovaries. For men, this includes radiation to the testicles or prostate. Many sexual side effects are caused by scar tissue from radiation therapy. Other problems, such as fatigue, pain, anxiety, or depression, can affect your interest in having sex.
Skin Changes
Radiation therapy causes skin cells to break down and die. When people get radiation almost every day, their skin cells do not have enough time to grow back between treatments. Skin changes can happen on any part of the body that gets radiation
Throat Changes
Radiation therapy to the neck or chest can cause throat changes because it not only kills cancer cells, but can also damage the healthy cells that line your throat. Your risk for throat changes depends on how much radiation you are getting, whether you are also having chemotherapy, and whether you use tobacco and alcohol while you are getting radiation therapy.
Urinary and Bladder Changes
Urinary and bladder problems may occur when people get radiation therapy to the prostate or bladder. Radiation therapy can harm the healthy cells of the bladder wall and urinary tract, which can cause inflammation, ulcers, and infection.
Source: National Cancer Institute, U.S. National Institutes of Health. www.cancer.gov

Role of radiotherapy in cancer management

Radiotherapy acts only on the irradiated tissues and can treat large areas of the body that may contain cancer. One dose of radiotherapy kills about half the cancer cells in the treated region. This powerful effect means that radiotherapy is also useful for palliation since low doses can result in significantly shrinking tumours with few side effects.

Curative radiotherapy

At least half of all patients for whom radiotherapy is prescribed, either alone or combined with surgery and/or chemotherapy, are treated with the goal of achieving a cure.

Radiotherapy is used by itself when it is the best treatment available because of the known cure rate or because it is likely to have fewer side effects. Examples include treatment for advanced cervix cancer, pituitary tumours, deep-seated gliomas and arterio-venous malformations, nasopharyngeal cancer and early stage low-grade lymphomas.

Because normal tissues recover from radiation damage better than tumours, it is possible to treat a cancer without completely destroying the host organ. Even if surgically removing the organ would lead to a more definite initial cure, provided radiotherapy is effective it is usually preferable so that the organ can be salvaged. Examples include larynx cancer and prostate cancer.

In the case of large tumours, radiotherapy is often combined with surgery to enable the whole tumour site to be treated without unduly affecting the patient's ability to function normally.

In general, radiotherapy is combined with surgery when:

- organ preservation is desirable -- for example, breast conservation treatment consisting of lumpectomy and radiotherapy
- the tumour is advanced with a high risk of local recurrence after surgery, for example, after a positive neck dissection or before surgery for rectal cancer
- an inoperable cancer can be rendered operable -- for example, fixed rectal cancers
- close surgical margins need treating to prevent local recurrence.

Chemotherapy may improve the results of radiotherapy through several mechanisms outlined in Table 1.

Table 1: Beneficial interactions between radiotherapy and chemotherapy.

Mechanism	Benefits	Examples of cancer
Spatial cooperation	Radiotherapy cures the high volume local cancer and chemotherapy cures micro-metastases.	Hodgkin's lymphoma Rectal cancer
Independent toxicity	Because radiotherapy and chemotherapy have different dose limiting toxicities, it is possible to deliver a higher anti-tumour dose with fewer side-effects than with radiotherapy alone.	Cervix cancer Oesophageal cancer
Enhanced tumour response	Even if the effects of radiotherapy and chemotherapy are only additive, the steep dose response of tumours means that there can be greater rates of cure than with radiotherapy alone.	Anal cancer

Palliative Radiotherapy

Incurable cancer causes many problems from local effects and the effects of spread to distant organs. Radiotherapy is the most effective treatment for people with incurable lung cancer. It can alleviate shortness of breath, cough and haemoptysis. It is also useful for patients with advanced and incurable cancers that are causing symptoms in the sites that they have arisen in. For breast cancer radiotherapy can control fungating masses, and for prostate cancer it can be used to relieve urinary obstruction.

Short course radiotherapy is an excellent treatment for the palliation of bone pain, brain metastases and compression of vital structures such as the spinal cord. In 80% of patients, a single dose of radiotherapy will relieve the pain caused by metastatic cancer in the bone. Radiotherapy is also very effective at relieving pain from compression of nerves. Radiotherapy can reverse the effects of spinal cord compression and prevent paraplegia.

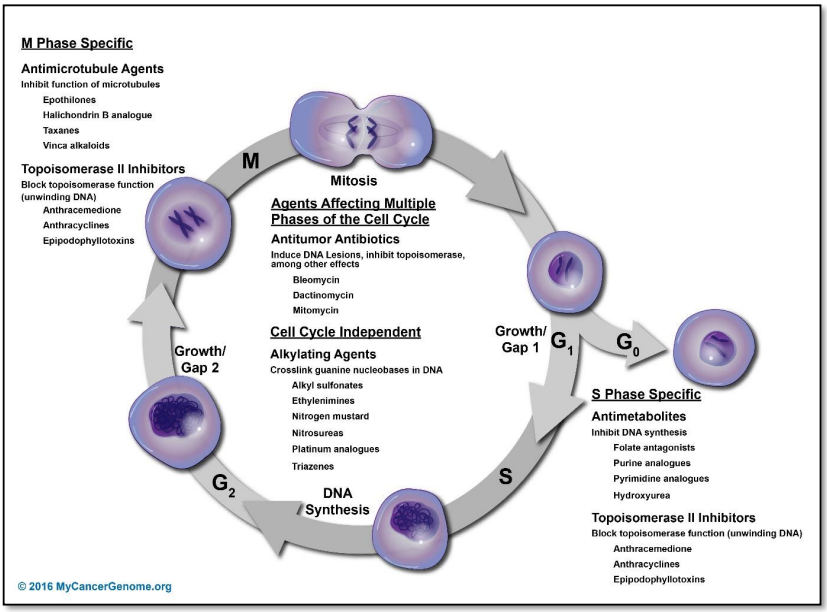
Radiotherapy may be used to prolong the life of patients with incurable cancers such as high-grade gliomas. Longer courses are required so that a sufficiently high dose may be given. Radiotherapy is more effective in these cases than chemotherapy alone.

Principles of Chemotherapy

Introduction

Medications in cancer care are utilized to achieve the following goals:

- to exert anti-cancer effect by killing or controlling cancer cells
- to treat complications arising from anti-cancer therapies
- to minimise the effects of cancer on body parts
- to control symptoms from cancer



Medications to achieve cancer cell death or cancer control

Chemotherapy, small molecules, monoclonal antibodies and hormonal agents are used for this purpose.

Chemotherapy

These agents achieve cancer control by several mechanisms. Some examples include:

1. damage to the DNA by various mechanisms during different phases of the cell cycle
2. inhibition of mitosis by interfering with microtubules.
3. To manipulate body immunity against cancer

Chemotherapy is used for the following purposes:

1. cure (e.g. leukaemia, lymphoma, germ cell tumours)
2. adjuvant, to eradicate micro metastasis following surgery or radiotherapy (e.g. breast, colon, ovarian cancers)
3. in combination with radiotherapy to sensitize radiotherapy (e.g. head and neck, rectum, lung, cervical cancers)
4. Palliation of symptoms and prolong survival (e.g. most metastatic cancers).

Small molecules

Many new medications inhibit various kinases along the signal transduction pathways that are associated with cell proliferation, survival, metastasis and angiogenesis.

Medication	Target Kinases	Current Use
Imatinib	Bcr-abl	Chronic myeloid leukaemia, gastrointestinal stromal tumour
Erlotinib,gefitinib	EGFR	Non-small cell lung cancer
Lapatinib, Trastuzumab	Her 2 receptor	Breast cancer
Sunitinib	PDGFR, VEGFR, Kit	Renal cell cancer, GIST
Everolimus	mTOR	Breast, renal cell cancer

Monoclonal antibodies

Monoclonal antibodies cause cancer cell death by the following mechanisms: [2]

1. direct action of antibody (receptor blockade or agonist activity, delivery of a drug or cytotoxic agent) E.g. Trastuzumab in HER2 positive breast cancer
2. Complement dependant cytotoxicity
3. Antibody-dependent cellular cytotoxicity (ADCC) E.g. Rituximab binds to CD 20 on lymphoma cells and stimulate ADCC
4. Regulation of T cell function E.g. Ipilimumab

In cancers like melanoma, antigen presenting cells (APC) present tumour antigen to the T cell via MHC complex. When tumour antigen binds with T cell receptors (B7 on APC with B28 on T cells), this leads to T cell activation and release of inflammatory proteins. When intracellular CTLA4 molecule is expressed on the surface of the T cell, binding of CTLA4 to B7 turns off the T cell activation. Ipilimumab binds to the CTLA4 and inhibit the binding of CTLA4 and B7. This means the T cell remains activated. Ongoing release of inflammatory proteins is expected to exert anti-tumour activity.

5. Specific effect on tumour vasculature and stroma E.g. Bevacizumab binds to VEGFR on stromal tissues and inhibit angiogenesis within cancer deposits.

Hormonal agents

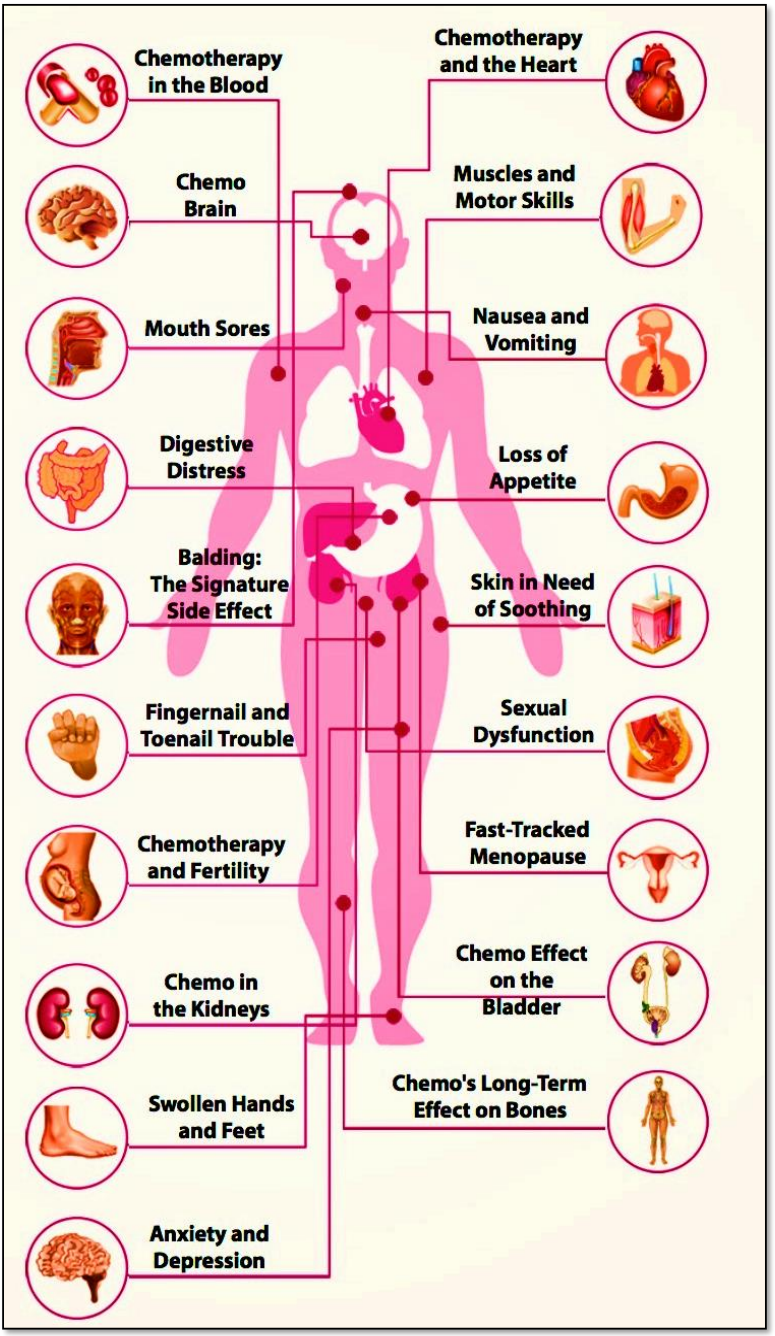
These agents inhibit the actions of various hormones on hormone responsive cancer cells. Side effects are usually related to hormone depletion.

Medication	How it works	Used in
Tamoxifen	Oestrogen receptor blockade	Breast cancer
Aromatase inhibitors	Inhibition of conversion of androgens to oestrogen	Breast cancer
Gonadotrophin releasing hormone(GnRH)	Decrease the ovarian production of oestrogen and testicular production of testosterone	Breast cancer and prostate cancer
Anti-androgens	Androgen receptor blockade	Prostate cancer

Managing chemotherapy side effects

Most side effects are due to non-specific damage to normal cells. They do not always occur and not all the patients experience the same side effects.

1. Nausea, vomiting -- Some drugs are more emetogenic than others; prophylactic antiemetic medications use can decrease the incidence.
2. Alopecia -- Not all the medications cause alopecia. Most breast cancer medications cause complete alopecia; most colorectal regimens do not cause complete alopecia.
3. Myelosuppression and neutropenic sepsis -- Neutropenic sepsis is treated with broad spectrum antibiotics. Period of neutropenia can be reduced by the use of granulocyte colony stimulating factors. Anaemia and thrombocytopenia can be treated with transfusions of packed red cells and platelets respectively.
4. Fatigue
5. Infertility -- Not all the drugs cause this side effect. It is important that pregnancy is avoided during and at least 12 months after completing chemotherapy. In ER negative cancers, ovarian suppression during chemotherapy can reduce the risk of infertility.
6. Effects on other organs -- E.g. cisplatin and neuropathy, anthracyclines and cardio toxicity, bleomycin and interstitial pneumonitis.


Chemotherapy in the Blood
Chemo Brain
Mouth Sores
Digestive Distress
Balding: The Signature Side Effect
Fingernail and Toenail Trouble
Chemotherapy and Fertility
Chemo in the Kidneys
Swollen Hands and Feet
Anxiety and Depression
Chemotherapy and the Heart
Muscles and Motor Skills
Nausea and Vomiting
Loss of Appetite
Skin in Need of Soothing
Sexual Dysfunction
Fast-Tracked Menopause
Chemo Effect on the Bladder
Chemo's Long-Term Effect on Bones

Medications to minimise the impact of cancer on organs

Examples:

Bisphosphonates (e.g. zelodronic acid, pamidronate) or RANKL inhibitor denosumab can decrease the incidence of skeletal related events in patients with bone metastasis from various cancers, including breast, prostate multiple myeloma, etc. Skeletal-related events include fracture, pain and hypercalcemia. These agents can also decrease the rate of osteoporosis/osteopenia caused by anti-hormonal agents like aromatase inhibitors.

Note: RANK is a surface receptor on pre-osteoclasts. RANK is activated by RANKL (RANK ligand) found on the surface of osteoblasts. Binding of RANK to RANKL leads to maturation of pre-osteoclasts to mature osteoclasts and subsequent destruction of the bones.

Managing symptoms caused by cancer or complications of treatments

When we attempt to treat cancers, it is important to make sure the quality of life of the patients is improved or maintained by controlling the cancer-related symptoms and treatment-related complications. Patients with metastatic disease can gain important benefits when their care is shared between various health professionals, including palliative care professionals.

Principles of Palliative Care

Introduction

Palliative care arose from the early Hospice Movement founded by Dame Cicely Saunders (1918-2005) in the UK. Dame Cicely was a nurse who re-trained as a medical social worker and eventually as a physician. She researched and championed effective pain management and promoted a more compassionate and holistic approach to patient care. Canadian urological surgical oncologist Dr Balfour Mount readily took up her reforms but because the word "hospice" did not translate well into the French-speaking province of Quebec, the term "palliative care" was adopted.

The World Health Organization has defined palliative care as "an approach that improves the quality of life of patients and their families facing the problem(s) associated with life-limiting illness, through the prevention and relief of suffering by means of early identification and impeccable assessment and treatment of pain and other problems—physical, psychosocial and spiritual" Currently, more than half of the world's countries have some form of palliative care services.

The principles of contemporary palliative care challenge a concept of cancer management that involves sometimes debilitating therapies and interventions, proceeding doggedly until the decision is made to "withdraw active treatment" and opt instead for comfort care and quality of life. Clinicians now opt for a more collaborative and cooperative management plan whereby symptom management, supportive care and amelioration of psychosocial, existential and spiritual concerns are acknowledged and managed conjointly between the primary care practitioners, oncologists and an interdisciplinary palliative care team. A patient may both appreciate and benefit from palliative care input early in the course of their cancer journey, and may prefer to negotiate a more gradual transition from active cancer treatments to management with primarily palliative care intent.

A reason for not discussing or offering palliative care in a timelier fashion is the concern that it will cause patients to lose hope. However, empathic and skillful communicators argue that it is possible to be honest without destroying all prospect of hope, and that the intent of treatment ought to be clearly presented to patients and their families.

The general public, and some health care practitioners, often misunderstand the contemporary definition of palliative care. There remains the mistaken belief that palliative care is only about the last few days or weeks of life. This belief is reinforced by the late referral of patients when they are seen to be "actively dying". To some extent, the limited number of palliative care services in Australia and New Zealand, and the limited resources and staff available to meet palliative care needs, exacerbates this situation.

When is it appropriate to request the involvement of palliative care?

Palliative care is a recognised specialty that is uniquely needs-based. Palliative care is not defined by a diagnosis, organ or system (as in cardiology or urology), but rather by the fact that any patient with a potentially life-limiting illness may experience suffering. That suffering may arise from the pathophysiological effects of a cancer (or non-cancer) diagnosis, the realisation of their own mortality, or the adverse effects of a range of surgical, physical or pharmacological therapies. If a patient is suffering persistent or difficult-to-control symptoms or emotional distress, then consultation with a palliative care practitioner should be considered.

Palliative care practitioners have considerable expertise in managing pain, nausea and vomiting, dyspnoea, delirium and the spectrum of symptoms experienced by cancer patients. They work collaboratively in an interdisciplinary team comprising a range of providers from medical, nursing, allied health, spiritual and supportive disciplines. The focus of this care is to anticipate problems that might arise and to minimise the impact of the progressing illness so that patients can

experience the maximum function and comfort that is possible within the limits of their illness.

It is important to guard against the use of “palliative” as a label that defines a patient or that describes the extent of, or limitations to intended treatment. The spectrum of investigations and interventions consistent with a palliative approach is guided by goals and expectations of individual patients and family, and by accepted standards of health care rather than being bound by preconceptions of what is or is not "palliative".

Palliative care is very often for people with cancer, but is also applicable to, and provided for, patients with a range of non-malignant life-limiting conditions. Increasingly, palliative care is provided to cancer patients in parallel to active treatments such as surgery, chemotherapy or radiation therapy and there is emerging evidence that the “parallel model of palliative care” improves patient and family comfort.

Despite recent advances in the prevention, detection and treatment of malignant disease, cancer remains one of the most dreaded diseases of our age. The role of the healthcare professional is not only to treat, but also to support, guide and empower the patient along their cancer journey. The national code of conduct for all doctors registered to practice medicine in Australia imposes a professional obligation to provide or arrange appropriate palliative care for all patients who need it or request it.

FAQs on Cancer

FAQ's on Cancer

1. What is cancer?

   Cancer describes a diverse group of diseases in which some of the body's cells become abnormal and begin to multiply out of control as a result of changes (mutations) in the genetic information of the cell. These abnormal cells can form an invasive (i.e. malignant) tumor which can invade and damage the area around them and spread to other parts of the body through the bloodstream or the lymphatic system.

2. What is primary and secondary site in cancer?

   The original site in which a cancerous tumour is formed is referred to as the primary site. The spread of cancerous cells from the primary site to another (i.e. secondary) site is referred to as metastasis.

3. How does Cancer start?

   The exact mechanism of transformation of normal cells to become cancerous is still not known. However, several factors are known to produce cancer, like tobacco in the form of smoking and chewing, alcohol, radiation, asbestos, certain chemicals, excessive fat consumption, viruses etc.

4. Does a lump always mean Cancer?

   No. A lump can occur in any part of the body which may or may not contain cancer cells. It could be benign like fibroadenoma in breast or malignant like carcinoma breast

5. What is the difference between a malignant and a benign tumour?

   Tumors can be benign (not a cancer) or malignant (a cancer). Benign tumours do not invade other tissues or spread to other parts of the body, although they can expand to interfere with neighbouring organs. The main features of a malignant tumour (cancer) are its

ability to grow in an uncontrolled way and to invade and spread to other parts of the body.

6. What are the most common Cancers seen in Indians?

Cancer affects different body organs due to a variety of factors and in India there are variations in the occurrence and pattern of cancer. However, in our men, the majority of all cancers are found in mouth and throat, lung, stomach,etc. and in women, uterus [cervix], breast, mouth and throat.

7. Does Tobacco Chewing or Smoking cause Cancer?

Smoking cigarettes is the cause of 75% of all lung cancers and 25% of all cancers in the Western countries. Smokers are also likely to develop cancer of the mouth, pharynx, esophagus [Gullet] pancreas and urinary bladder. In our country, almost 50% of all cancer in men are caused by heavy tobacco chewing and smoking; rather it is worse. The "bidi" contains more toxic harmful chemicals than in cigarette which produces cancer. Tobacco smoking is also the major cause of heart disease, gastric ulcers, chronic bronchitis, and emphysema, often the smoker dies young.

Due to the husband's smoking habits, wife [**passive smoking**] is also at a high risk for developing cancer of various organs.

8. Does heavy Alcohol consumption cause Cancer?

Heavy alcohol consumption especially in the presence of smoking increases the risk of certain cancers like cancer of oral cavity, stomach and liver.

9. Is Cancer Hereditary?

Certain cancers are seen to occur in families. Eg Breast, colorectal cancer. But less than 5 % of cancers are hereditary.

10. Do Viruses cause Cancer?

There is good evidence to suspect that viruses may cause cancer in humans - leukemia, nasopharyngeal cancer, liver and cervix cancer are considered now as being caused by viruses.

11. Is Radiation a Cause?

Radiation is widely used to diagnose many diseases. Radiation is also a very successful method to treat cancer. These are absolutely proper and safe uses. However Radiation is extremely dangerous and can cause cancer when not properly handled. Over exposure to X-rays, or byproducts of atomic radiation can contribute to the development of cancer especially leukemia.

12. How is Cancer Diagnosed?

Diagnosis is established by history, clinical examination, imaging and biopsy.

13. Is it possible to prevent cancer?

It is recognized that some cancers can be prevented. Most mouth cancers can be prevented by not chewing tobacco and most lung and throat cancers can be prevented by not smoking. Certain cancers are produced by chemicals used in industry. If workers are protected against contact with such occupational factors, some cancers can be prevented.

A well balanced diet with less fat and more green leafy vegetables, citrus fruits, milk etc.can protect people. Above all, even though not all cancers are preventable, many deaths due to cancer can be prevented by early detection and proper treatment. A knowledge of the warning signals of cancer helps one to get an early diagnosis and a successful treatment.

14. What is the best method by which cancer is treated now?

The best treatment method for cancer depends on several factors, of which the most important are the site of cancer, type of cancer, spread of the disease when diagnosed, general health of the patient

etc. Surgery, chemotherapy and radiotherapy are the proven modalities of treatment, which may be used alone or in combination.

15. How is an early cancer detected?

One should know the warning signals of cancer. These need not always be due to cancer, but when a symptom is noted, promptly undergo a through medical examination by a competent physician or cancer specialist. Secondly, self-examination methods and periodic self-examination of mouth and breast can detect many cancers in early or precancerous stages. In case of uterine cervix a simple Paps test can help in early detection.

16. How does early diagnosis help?

When cancer has not spread beyond its site of origin a complete cure is often possible. When a cervix cancer is localized, almost 100% cure is possible but when it has spread to nearby bladder or rectum, not even 20% will live up to 5 years. Similarly, an early breast cancer when treated adequately, would have a more than 85% chance of surviving more than 5 years whereas when it is spread to lungs less than 10% would survive 5 years. Thus early diagnosis ensure a better cure rate.

No. Pain is not a symptom of early cancer, it is a symptom of advanced cancer.

17. What are the warning signals of cancer?

- A sore or ulcer that does not heal within 3 weeks with antibiotics especially in mouth. Pearly white patches in the mouth.
- Unusual bleeding or discharge from any orifice.
- Thickening or lump in breast or elsewhere in the body.
- Repeated abdominal cramps, indigestion. Difficulty in swallowing which lasts more than 3 weeks. Changes in bowel habits [Diarrhea & Consumption] lasting for more than 3 weeks.

- Hoarseness of voice, nagging cough; persisting even after treatment for 3 weeks.
- Difficulty in urination, blood in urine, especially above 45 years of age.
- Change in wart or mole.
- Listlessness, anemia, intermittent fever which are not controlled by medication.
- **These signs need not always be due to Cancer. But consult a specialist doctor if you have any of the above symptoms**

18. Is cancer contagious? Can the person affected with cancer can spread the disease to other family members during close contact?

Cancer is not a contagious disease and there is no bar on maintaining normal contacts with other persons including children. This disease is not transmitted from one person to the other by close contact including sharing utensils, clothes, food, drinks and sexual activity.

19. How to perform Visual Inspection with Acetic acid (VIA) for screening of cervical cancer?

VIA involves naked eye examination (without magnification) of the uterine cervix after application of freshly prepared 3-5% acetic acid (vinegar) for a minute, under a good light source. The test results are scored as positive when a well-defined, dense acetowhite area with regular margins appears near to the squamo columnar junction.

20. What are the symptoms of cervical cancer?

- **Vaginal bleeding:** This includes bleeding between periods, after sexual intercourse (post coital) or post-menopausal bleeding.
- **Unusual vaginal discharge:** A watery, pink or foul-smelling discharge is common.
- **Pelvic pain:** Pain during intercourse or at other times may be a sign of abnormal changes to the cervix.

All of these cervical cancer symptoms should be discussed with the doctor.

21. Can Benign Cysts Become Cancerous?

Cysts are very common in breasts and rarely turn into cancers. But it's extremely important to find out whether, what you have is just a cyst or something else.

22. What three steps are very vital for evaluation of any breast lump?

- Clinical breast exam
- Mammogram
- FNAC

23. Is areca nut, masheri harmful?

Yes they cause cancer. Also they predispose to pre-cancerous conditions like erythroplakia, leuoplakia and submucous fibrosis (difficulty opening mouth)

24. Does vaccination reduce risk of cancer?

Yes. It helps in reducing liver cancer. Also there are vaccines to prevent cervix cancer, but its effect is yet to be proven.

25. What are the ideal screening investigation for females?

For early diagnosis of cancer cervix, Pap `s smear is done starting from 35 years of age, once in three years is recommended

For early diagnosis of breast cancer, Mammography is done starting from 50 years of age, once in 18 – 36 months is recommended.

26. What is chemotherapy?

Chemotherapy refers to certain drugs, which can inhibit or kill cancer cells. It can be given in the form of injections or oral medications. Chemotherapy is usually administered in 'cycles'- repetitively every few weeks.

27. How is it given?

Each cycle is given over one or several days. The cycle duration (time interval between two cycles, counting from the 1st day of initial cycle) and number of cycles varies between different tumors and different regimens. The majority of solid tumours (e.g. lung, breast, ovarian, gastro-intestinal and other similar tumours) are treated with 4 to 8 cycles (average 6) of chemotherapy administered every 3 to 4 weeks

28. What is the purpose of chemotherapy treatment?

The aim of giving chemotherapy depends on the stage of cancer being treated. In early stage disease it is given to prevent relapses. In advanced stage, the purpose is to control symptoms, improve the quality of life and in many instances improve survival.

29. What are the common side effects of chemotherapy?

The relatively common side effects and toxicities of chemotherapy include, but are not restricted to the following

a) Low white blood cell counts (leukopenia) with or without infection
b) Low platelet counts ,Anemia
c) Nausea and/or vomiting:
d) Sores and ulcers in the mouth and throat:
e) Diarrhoea: Constipation:
f) Hair Loss:
g) Skin toxicity:
h) Pain and aches in limbs and other parts of the body
i) Organ failure

30. What does palliative care mean?

Palliative care or best supportive care is offered to patients who suffer from final stages of cancer, where curative treatment is not possible. It aims at improving quality of life.

31. Does alternate medicine help in curing cancer?

Only proven treatment for cancer is surgery, Radiation and allopathic drugs (chemotherapy). Hence role of alternative medicines in treatment of cancer is questionable.

31. Is the hair loss permanent after treatment?

No. It is temporary and regrows after couple of months of stopping treatment.

32. Will the patient loose fertility after treatment?

Possibility of loosing hair is high with increasing age. There is always option of preserving patient`s sperm or ovum for future, if they decide to bear children.

33. Can cancer come back once cured?

There is a risk that some cancer can recur. It depends on the advanced stage of the disease at diagnosis, type of cancer and adequacy of treatment. Hence, adequate follow up is necessary.